

COMPENG 4TL4- Lab 5 Report: FIR and IIR Filter Design and Analysis

Student Name	Student Number
Kevin Le	400385350
Jonathan D'Souza	400304744

1. The truncation method of FIR design

This section details the design and analysis of linear-phase lowpass FIR filters using the truncation method, comparing the results with the `firls()` function and analyzing the effects of convolution and the `filter()` function.

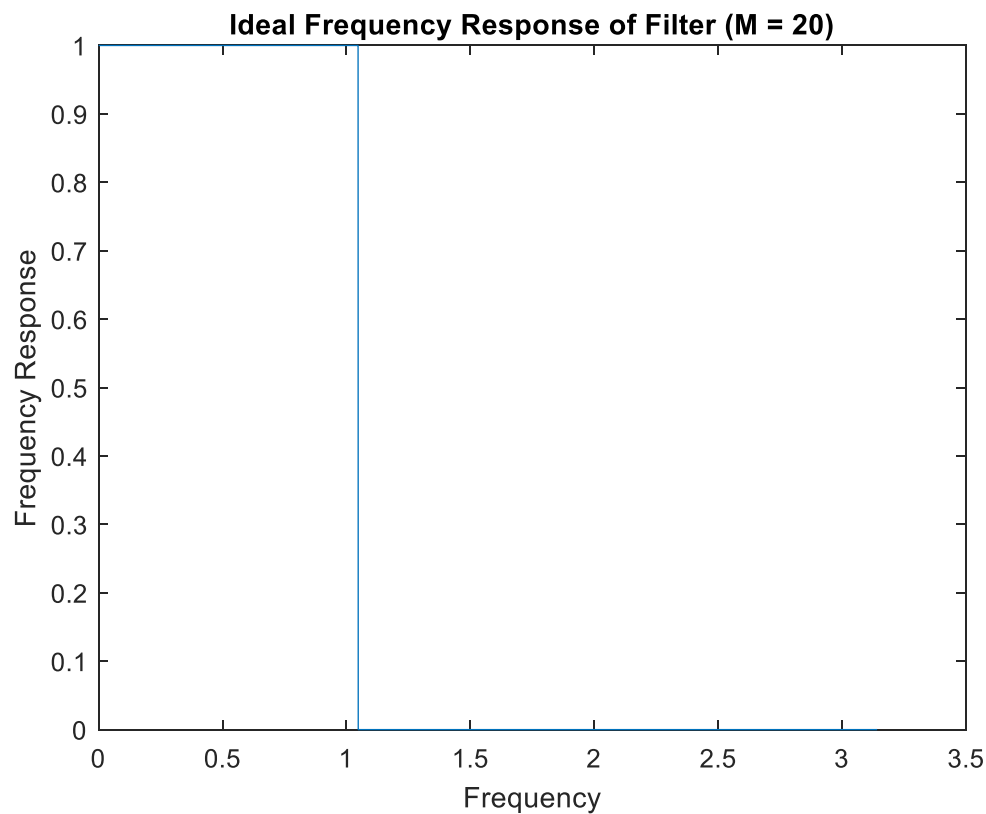


Figure 1: Magnitude plot of the ideal filter frequency response, with cutoff frequency $\pi/3$

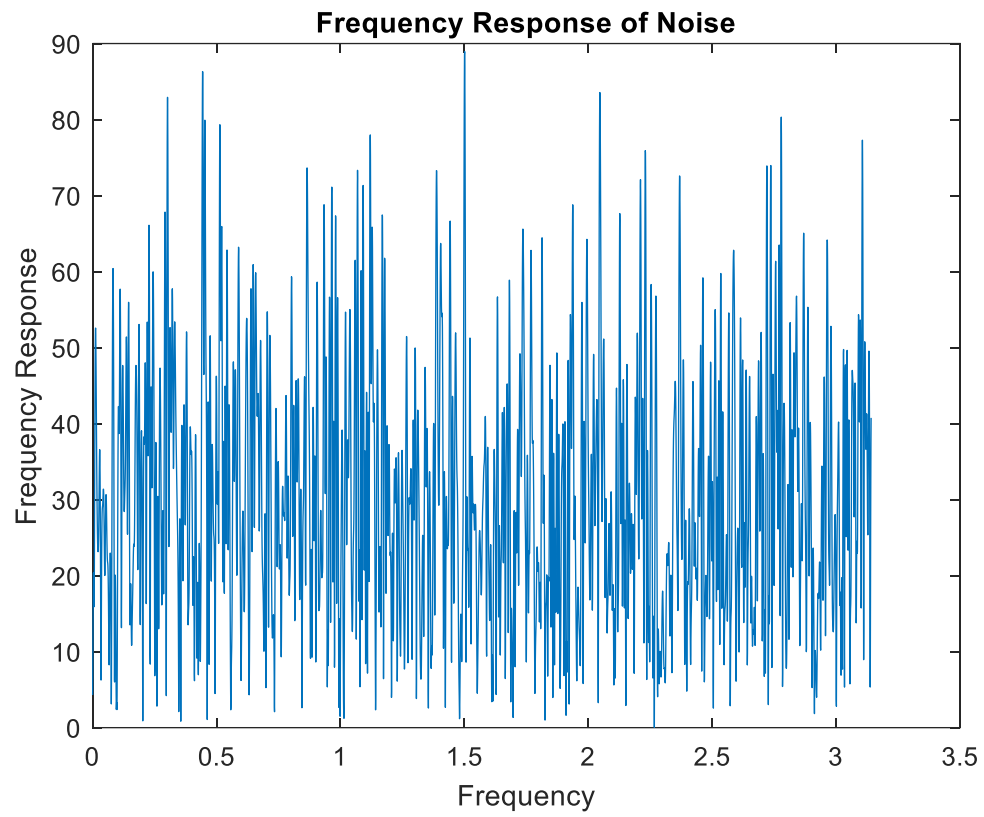


Figure 2: Magnitude of frequency response of noise before filtering

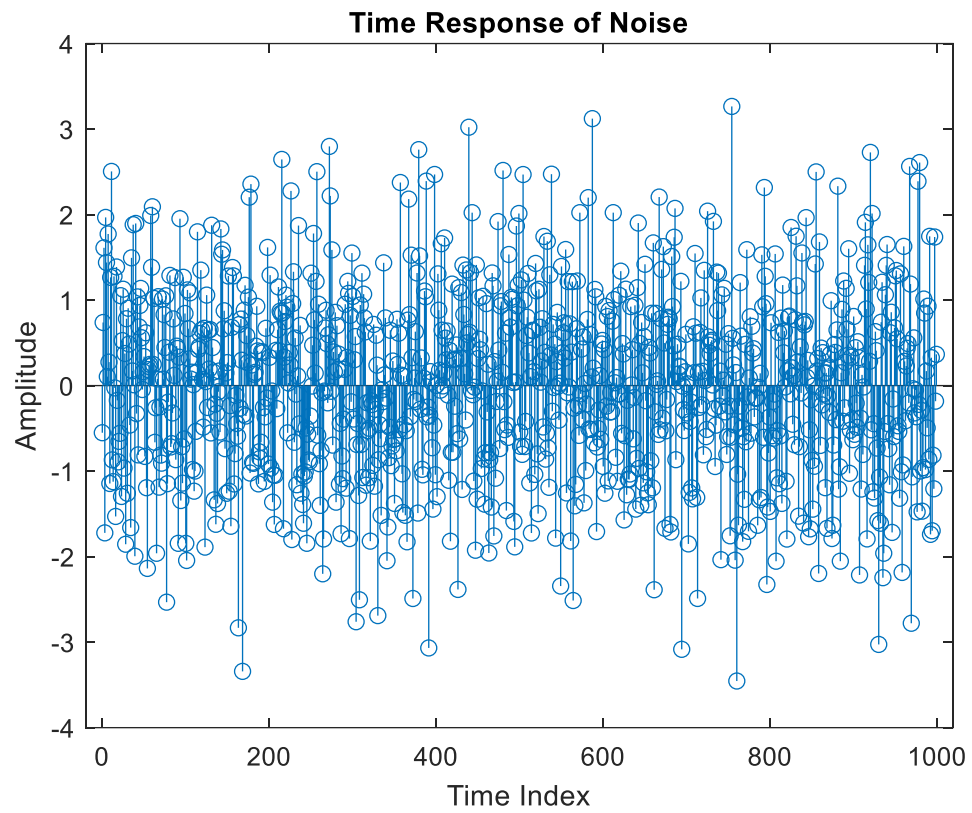


Figure 3: Noise in time domain before filtering

Filter Order $M=20$

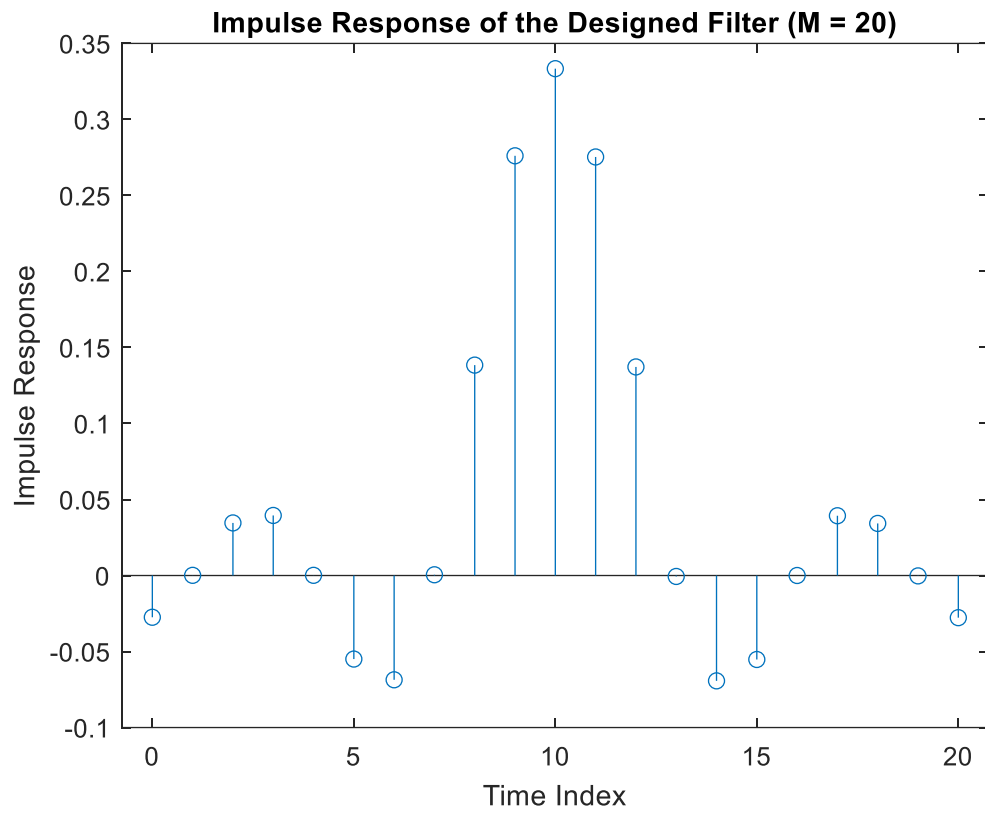


Figure 4: Impulse response of truncated filter with order $M=20$

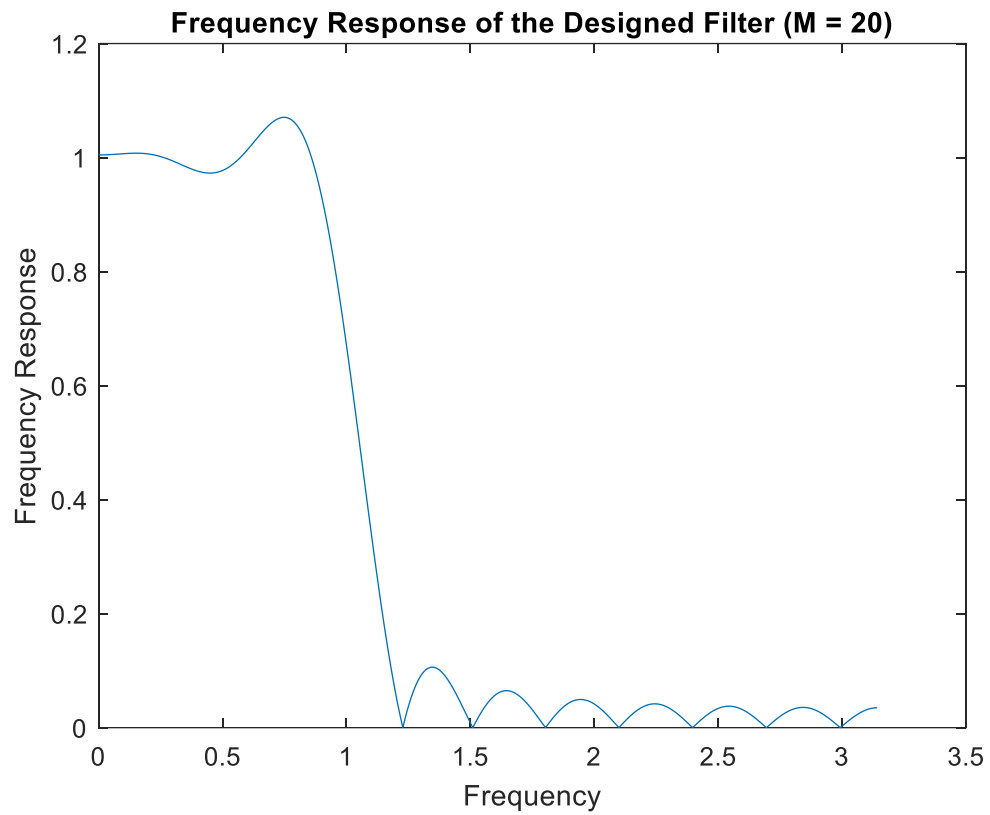


Figure 5: Magnitude of frequency response of truncated filter with order $M=20$

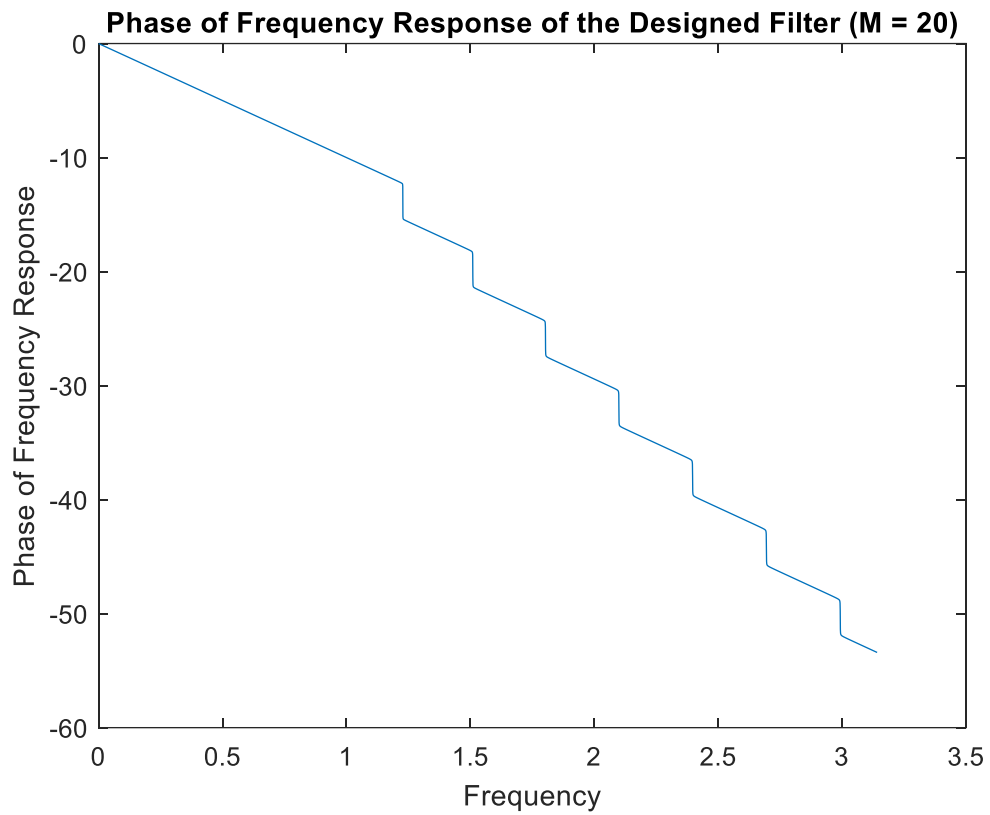


Figure 6: Phase of frequency response of truncated filter with order $M=20$

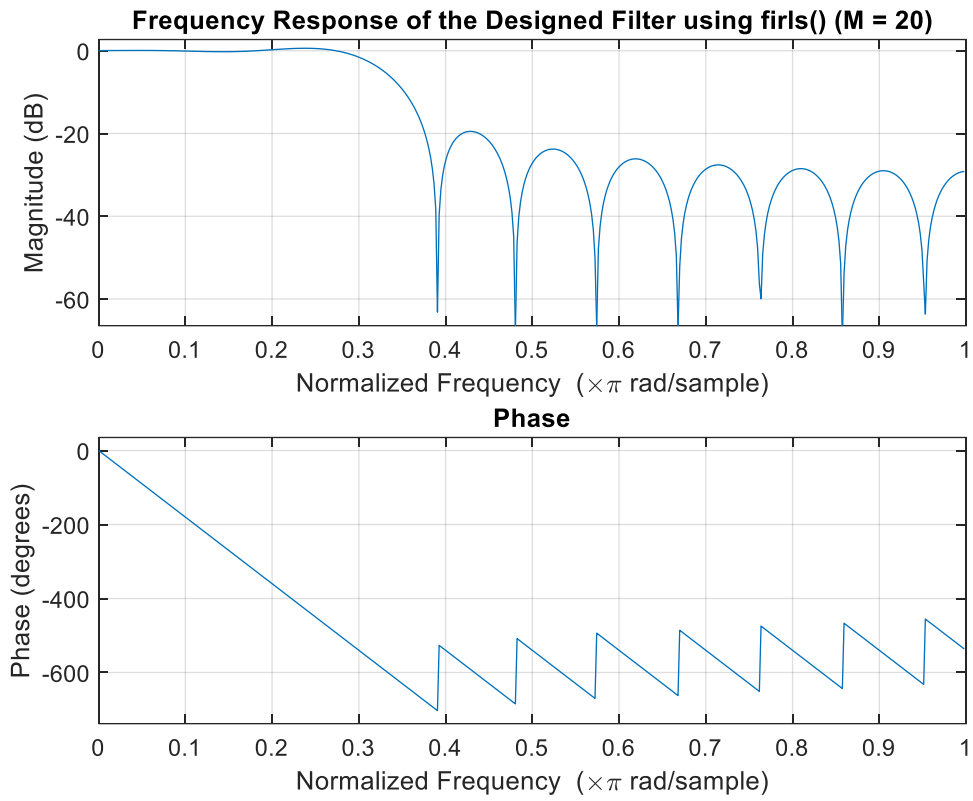


Figure 7: Frequency response of truncated filter with order M=20 using firls()

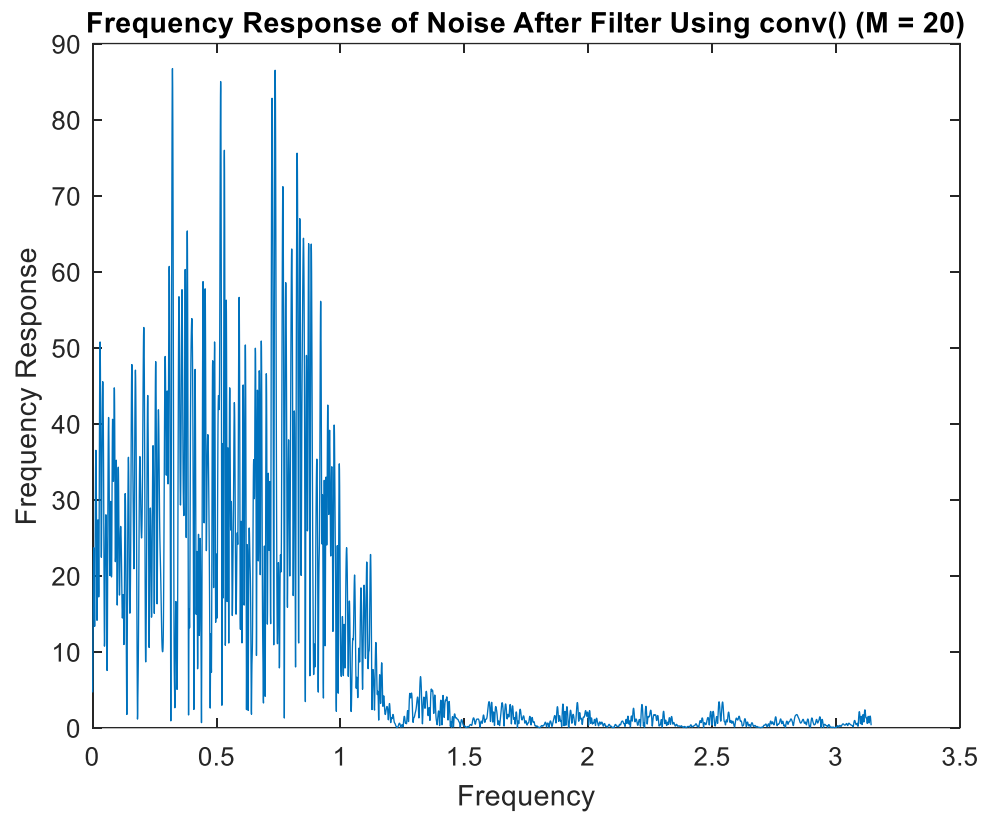


Figure 8: Magnitude of frequency response of noise after applying filter with order $M=20$ using conv()

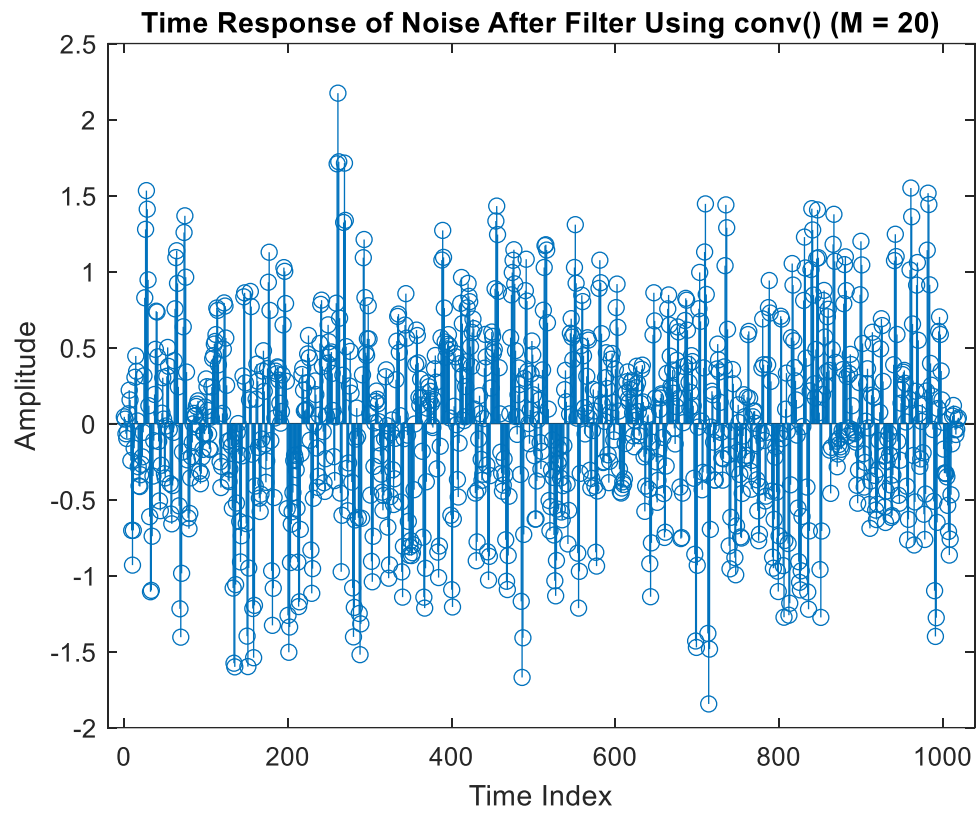


Figure 9: Time response of noise after applying filter with order $M=20$ using conv()

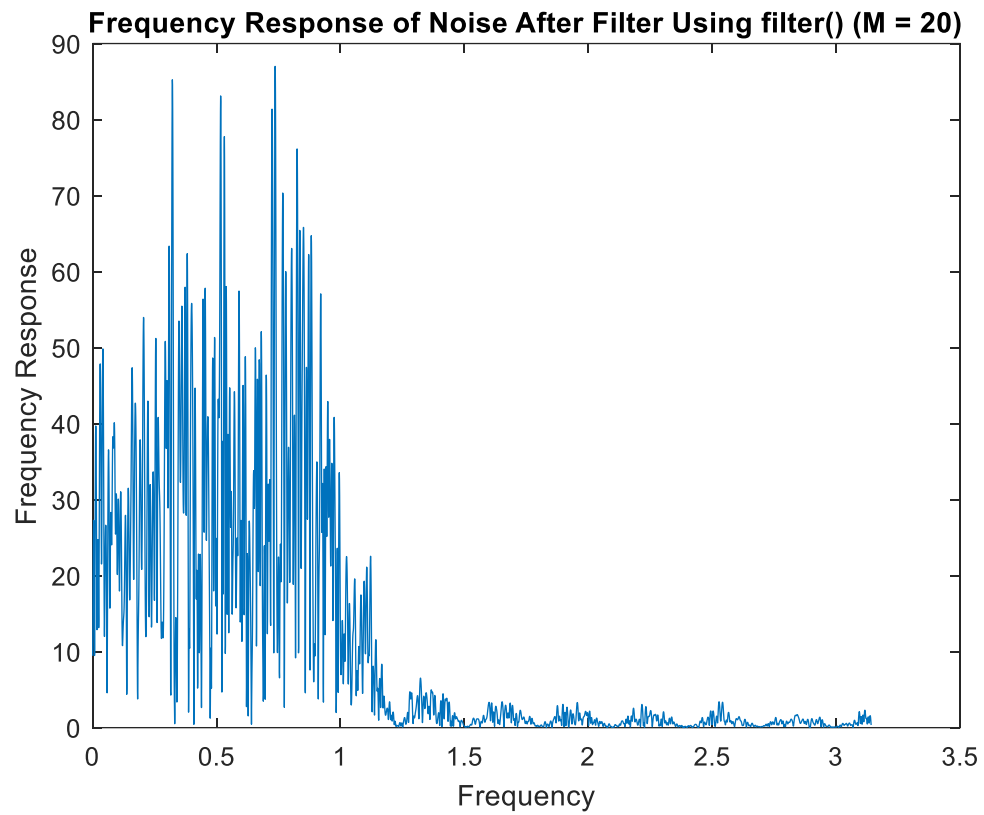


Figure 10: Magnitude of frequency response of noise after applying filter with order $M=20$ using filter()

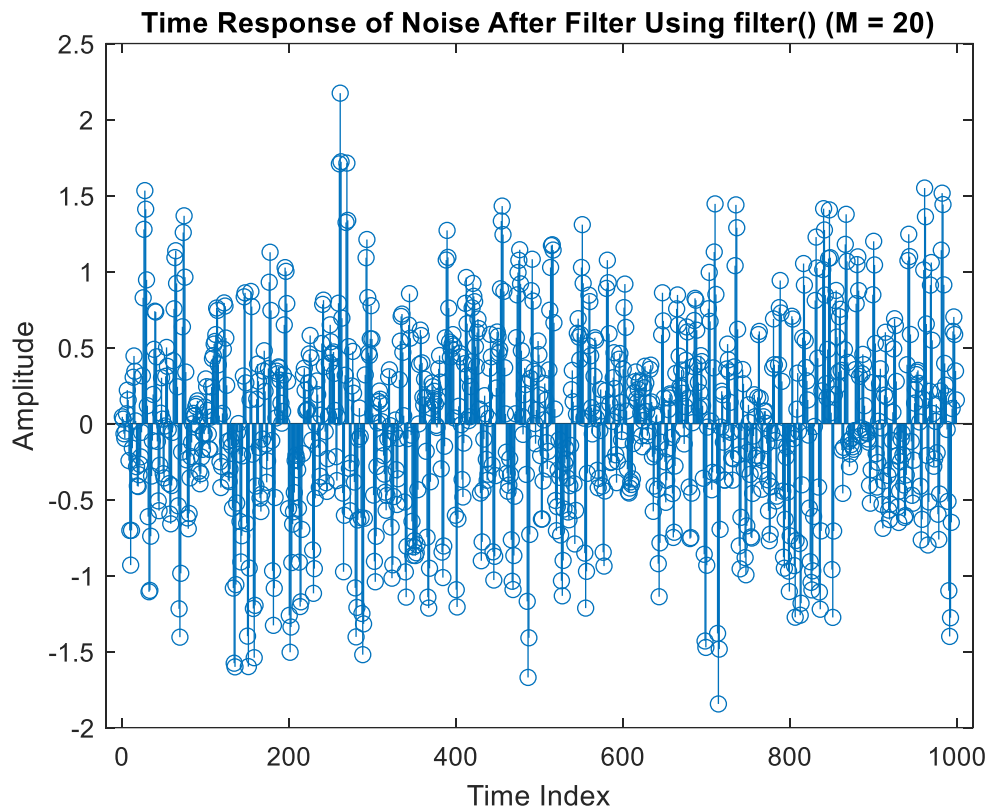


Figure 11: Time response of noise after applying filter with order M=20 using filter()

Filter Order M=50

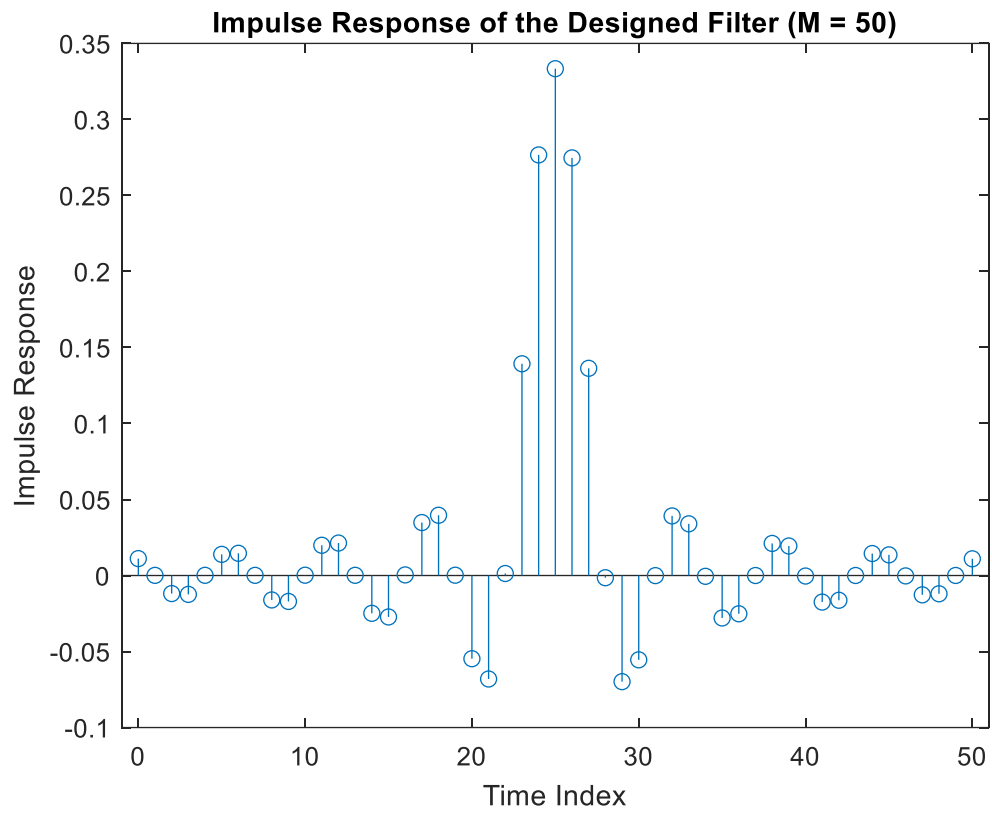


Figure 12: Impulse response of truncated filter with order M=50

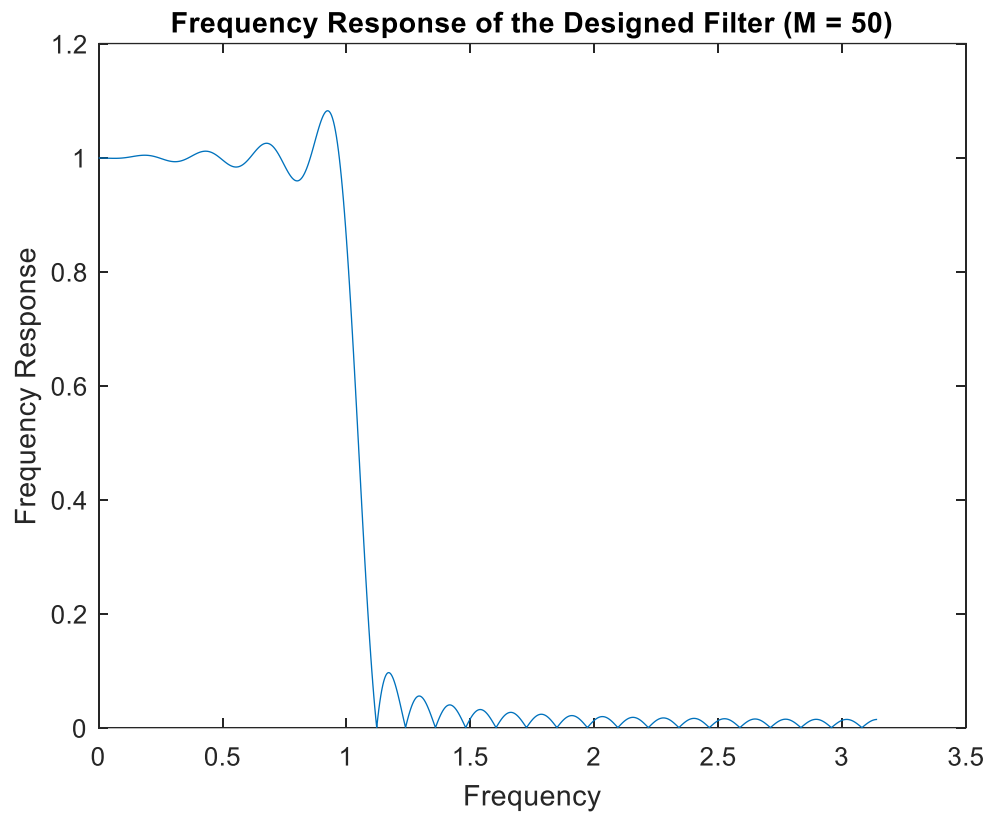


Figure 13: Magnitude of frequency response of truncated filter with order $M=50$

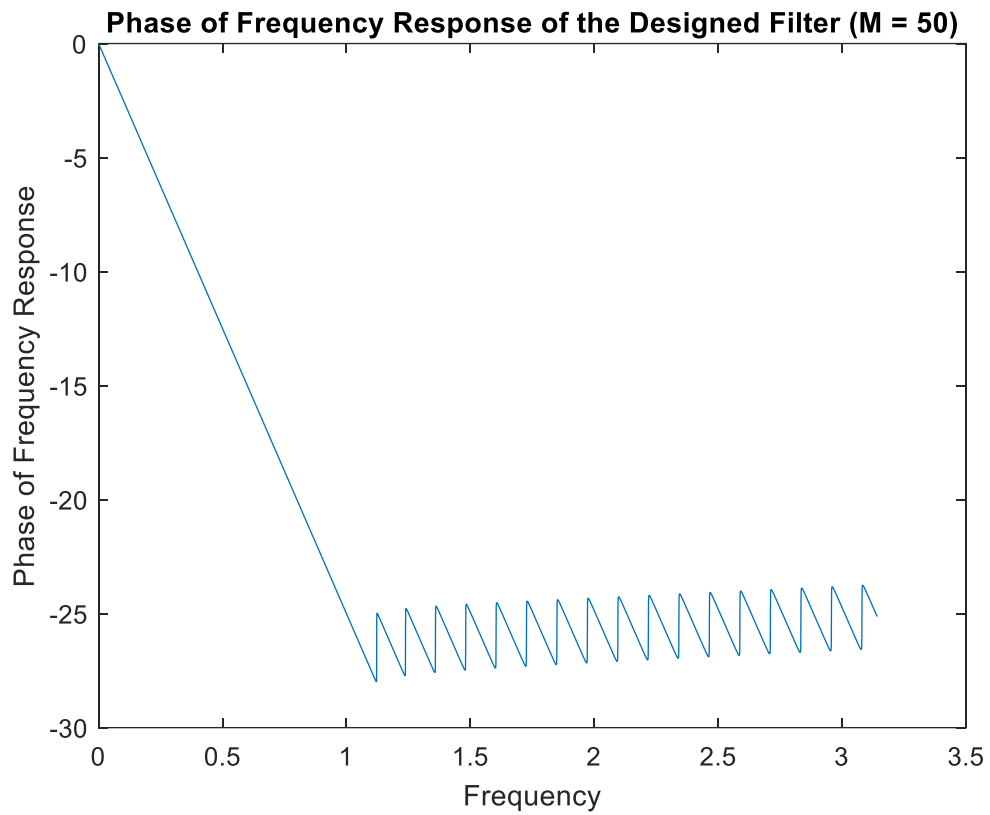


Figure 14: Phase of frequency response of truncated filter with order $M=50$

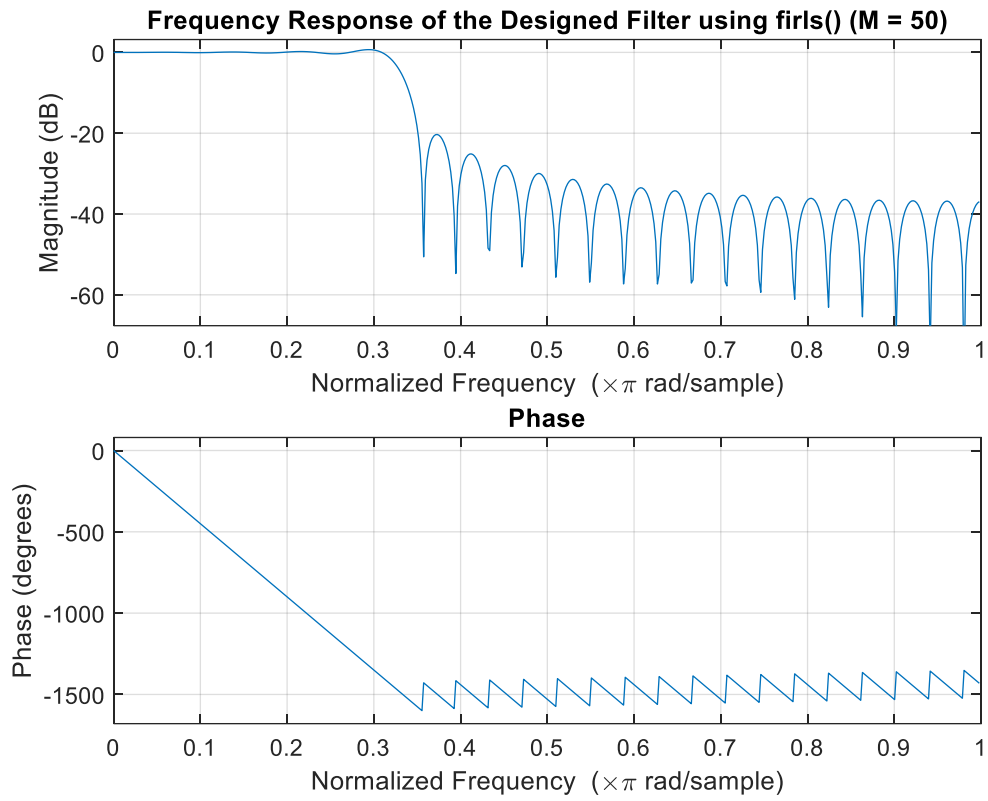


Figure 15: Frequency response of truncated filter with order M=50 using firls()

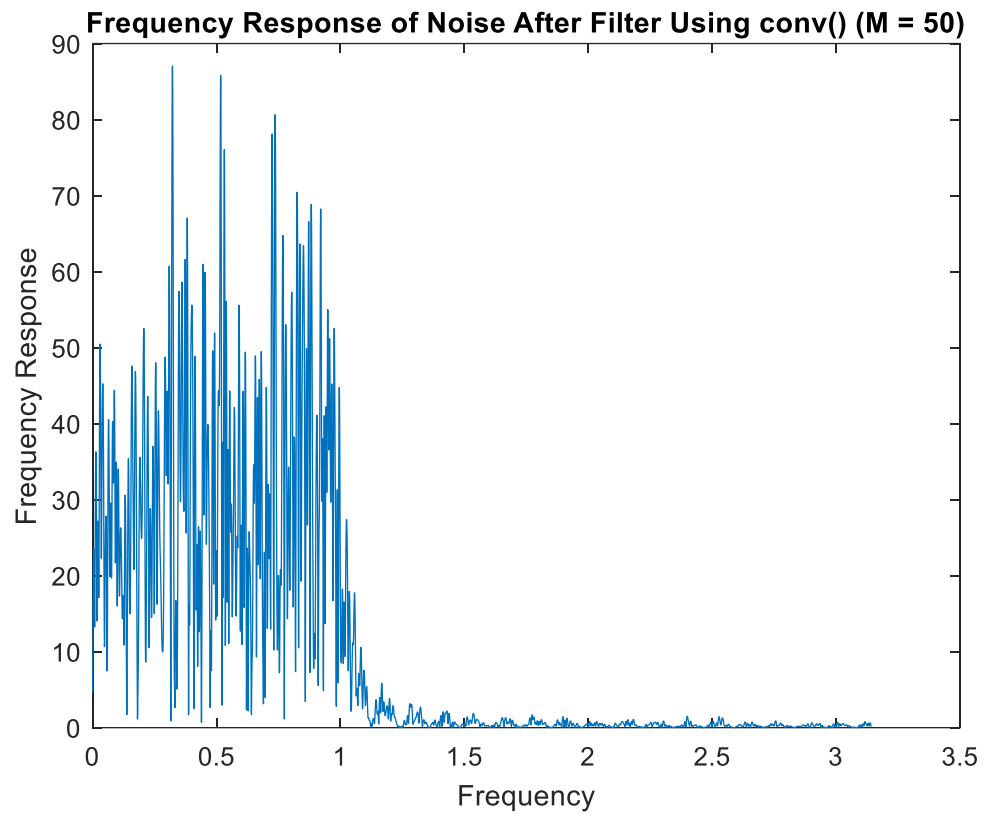


Figure 16: Magnitude of frequency response of noise after applying filter with order $M=50$ using conv()

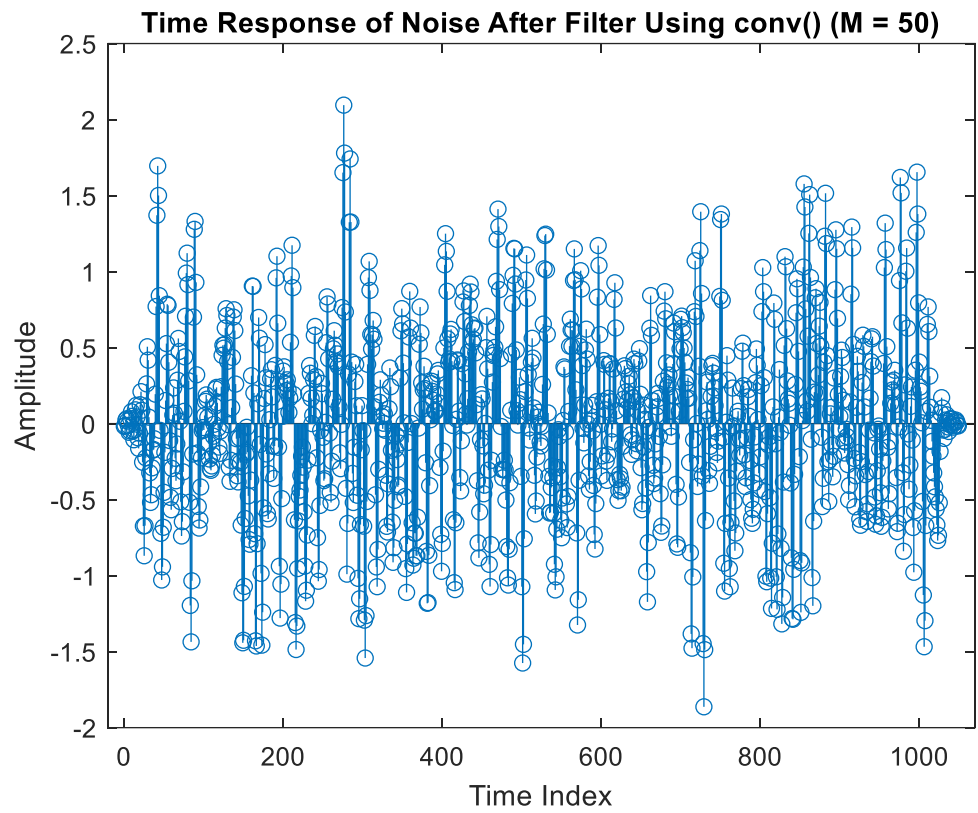


Figure 17: Time response of noise after applying filter with order M=50 using conv()

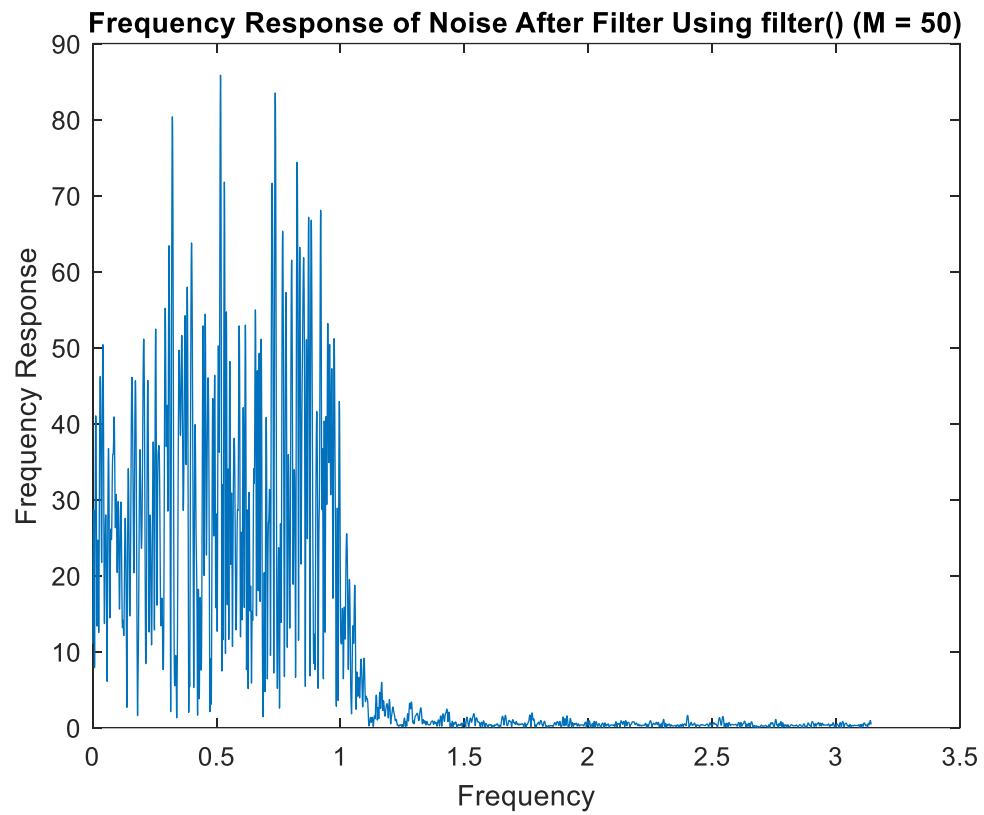


Figure 18: Magnitude of frequency response of noise after applying filter with order $M=50$ using filter()

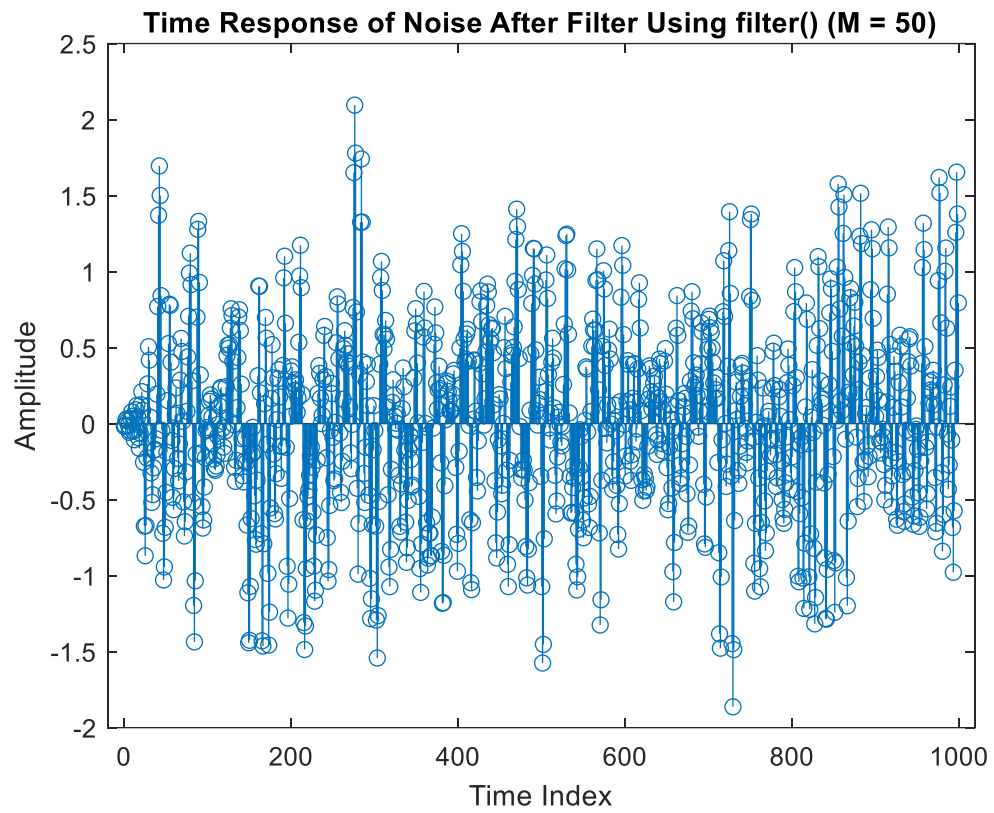


Figure 19: Time response of noise after applying filter with order M=50 using filter()

Filter Order $M=150$

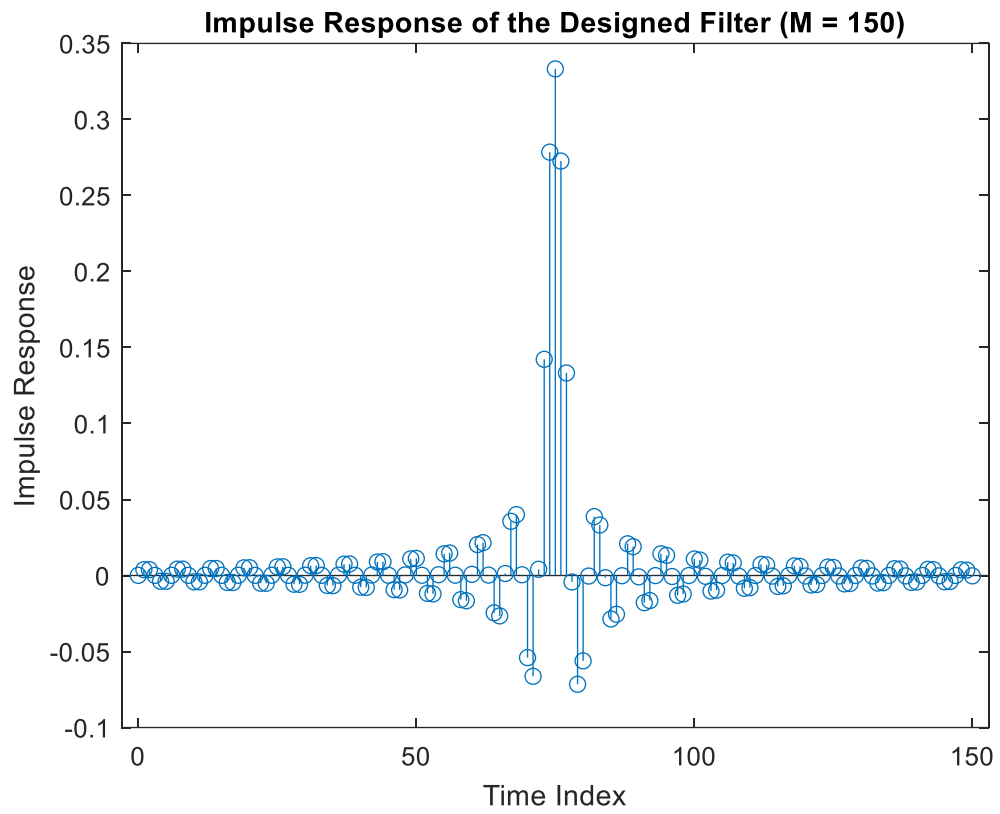


Figure 20: Impulse response of truncated filter with order $M=150$

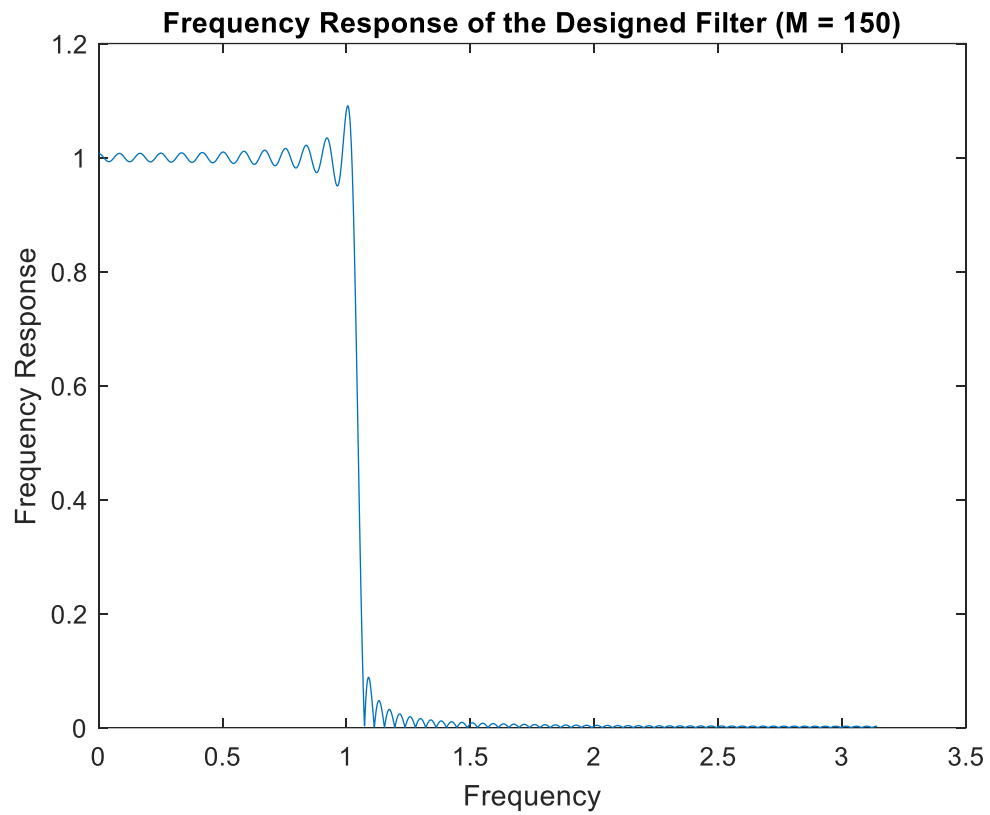


Figure 21: Magnitude of frequency response of truncated filter with order $M=150$

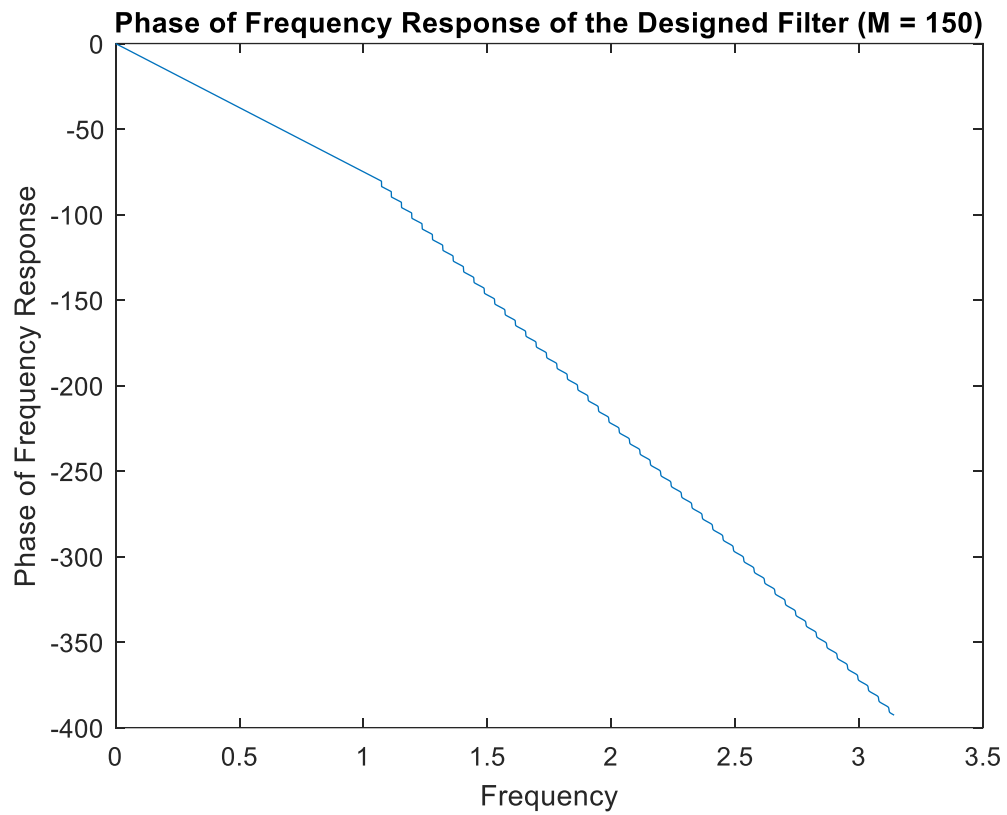


Figure 22: Phase of frequency response of truncated filter with order $M=150$

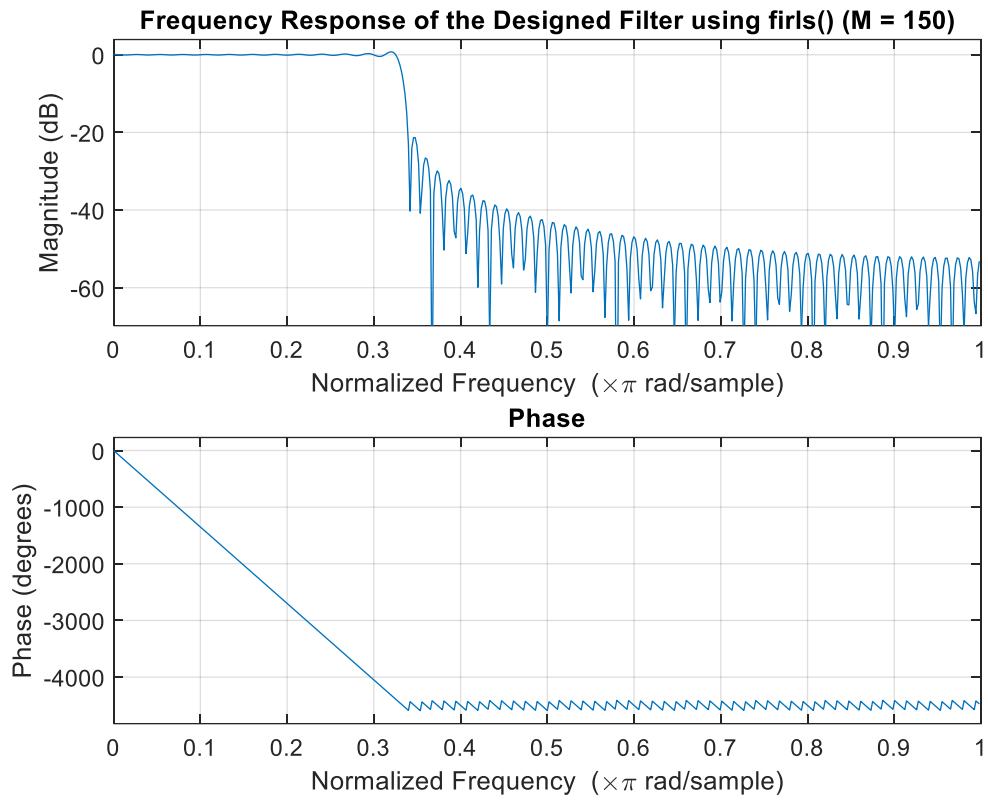


Figure 23: Frequency response of truncated filter with order M=150 using firls()

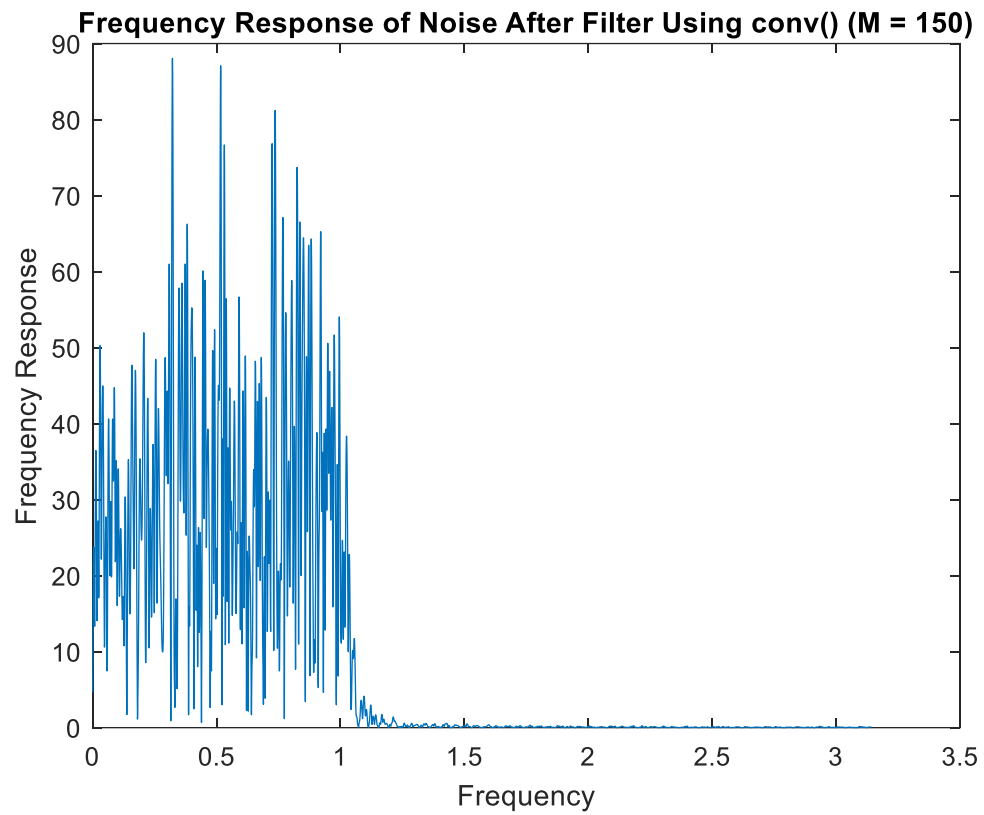


Figure 24: Magnitude of frequency response of noise after applying filter with order $M=150$ using conv()

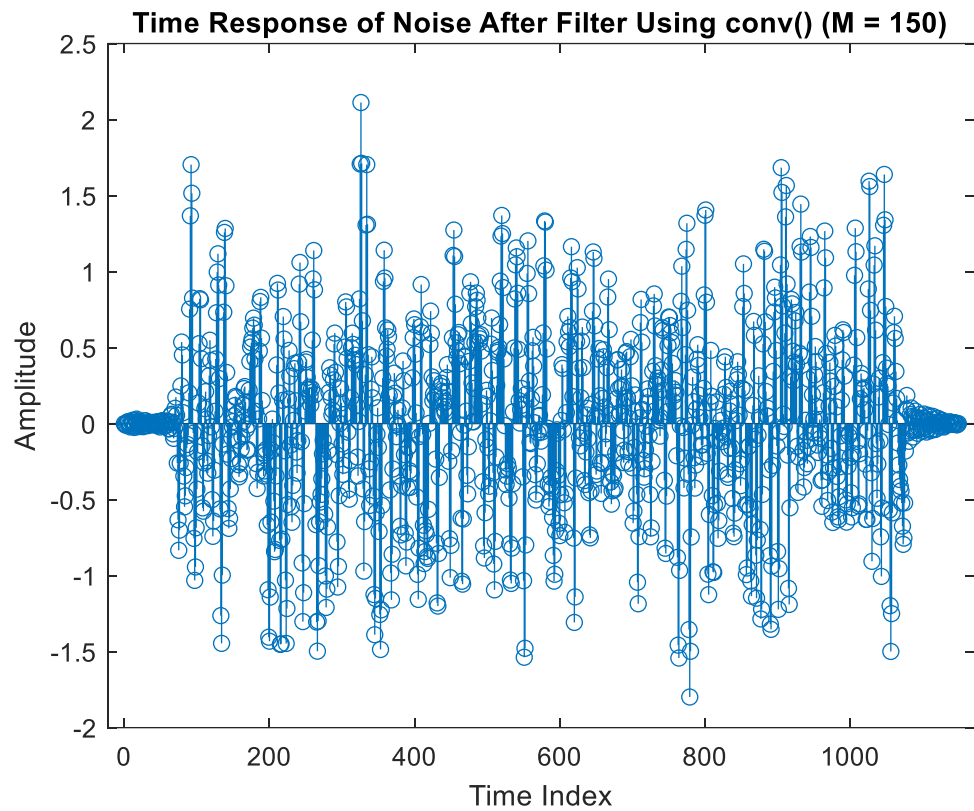


Figure 25: Time response of noise after applying filter with order M=150 using conv()

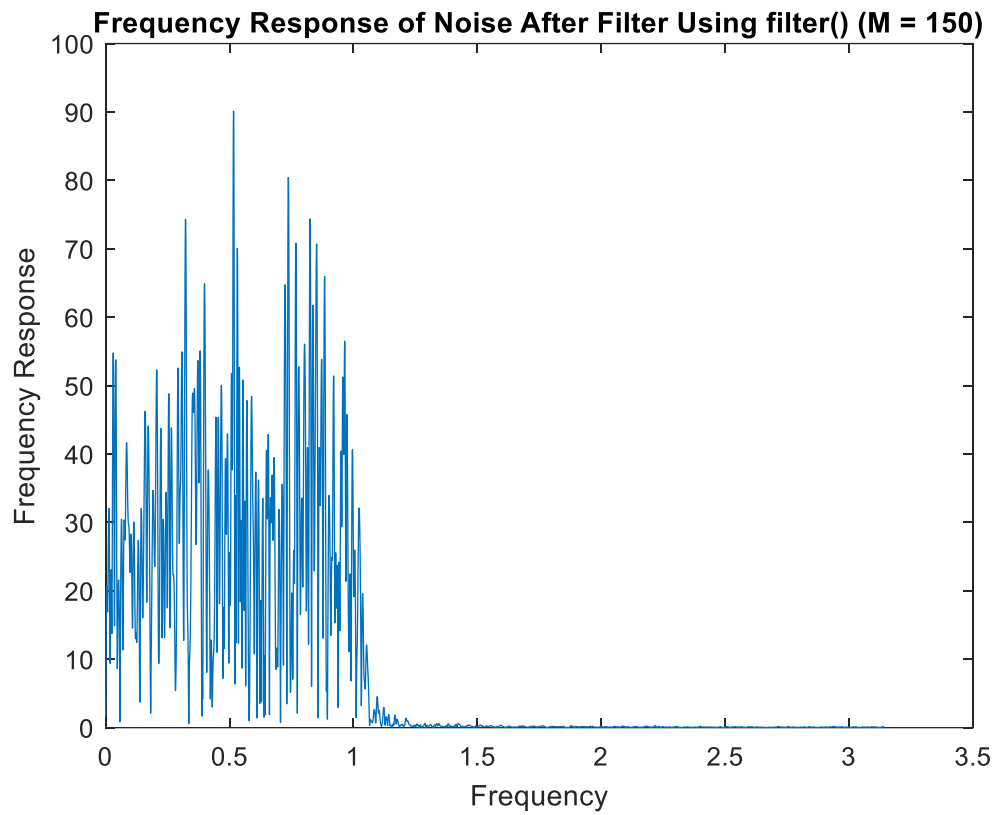


Figure 26: Magnitude of frequency response of noise after applying filter with order $M=150$ using filter()

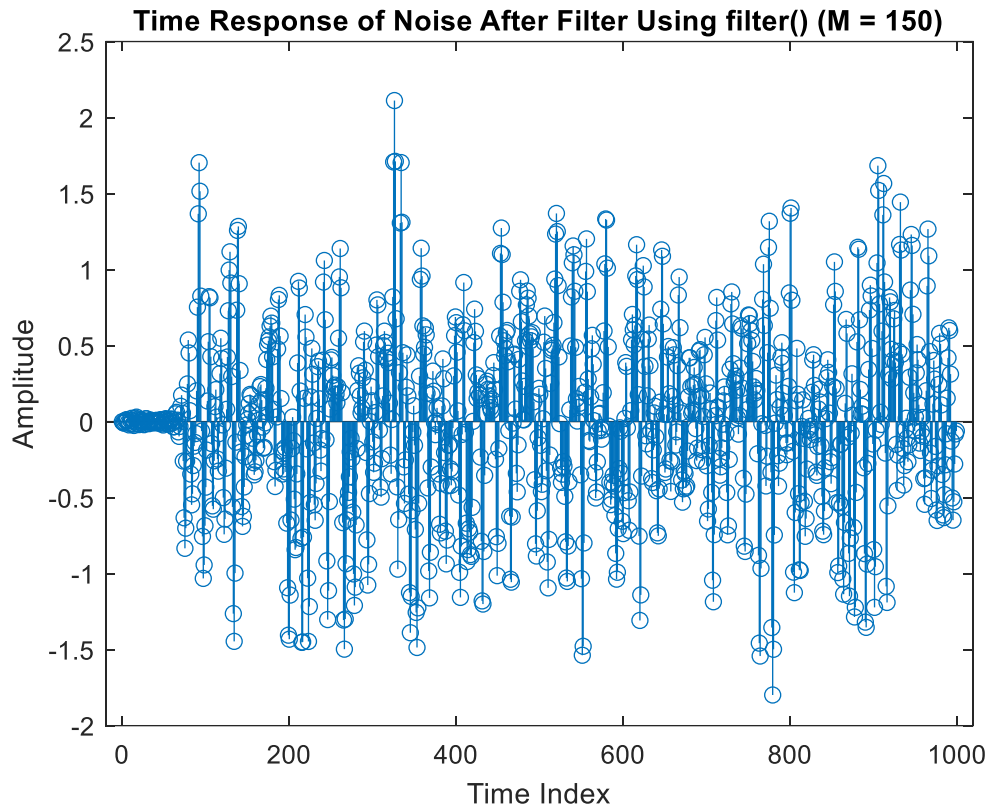


Figure 27: Time response of noise after applying filter with order $M=150$ using `filter()`

The major difference between using the `conv()` and the `filter()` function is the length of the signal in the time domain. The noise signal started with 1000 samples, but when filtered using `conv()`, the resulting signal has a length higher than 1000 samples. To be more exact, the length of the signal after filtering with `conv()` is $1000+M$. This is because the convolution operation returns a signal with length equal to the sum of the length of its two inputs minus one. On the other hand, when applying the filter using the `filter()` function, the resulting signal has length 1000, equal to the length of the original noise signal.

Besides the differences with the signal samples after 1000, the remaining signals from the first 1000 samples seem identical between the `conv()` and `filter()` functions. There is also almost no difference in the frequency domain signals when using `conv()` vs. `filter()`, and we can observe significant attenuation to frequency components after the cutoff frequency in both the signal using `conv()` and `filter()`, just as expected when using a low pass filter.

2. Chebyshev Type II lowpass IIR filter

2(a) Design linear-phase lowpass FIR filters with the cutoff frequency $\omega_c = \pi/3$ using the window method with a Dolph-Chebyshev window for filter orders $M=20, 50$ and 150 . The MATLAB Signal Processing Toolbox function `fir1()` can be used to implement the window method and the MATLAB Signal Processing Toolbox function `chebwin()` can be used to obtain the desired Dolph-Chebyshev windows. Note that you will need to specify the amount of sidelobe attenuation for your Dolph-Chebyshev windows. Try some different values and demonstrate what effect this has on: i. the

passband ripple and stopband attenuation. ii. the slope of the transition region. for one of the filters (M=20, 50 or 150).

Answer and Discussion:

These filters were designed using attenuation of 25, 50, and 100 on filter order M=20. Looking at the frequency responses of Figure 28, 29, 30, we can observe that with increasing sidelobe attenuation there is decreased ripple in the stop band. The pass band ripple cannot really be seen but it is important to note that the pass band gets extended as the sidelobe attenuation increases. This is not ideal behavior as we want maximum attenuation right after our cutoff frequency is reached. The slope of the transition region increasing with sidelobe attenuation is bad for our filter design although it does decrease ripples. We see that with increasing filter order in Figures 30, 31, and 32; we get a steeper cutoff at the cost of having more ripples. Finally, we see that this behavior in frequency response for both changing attenuation and order can also be seen in the phase response.

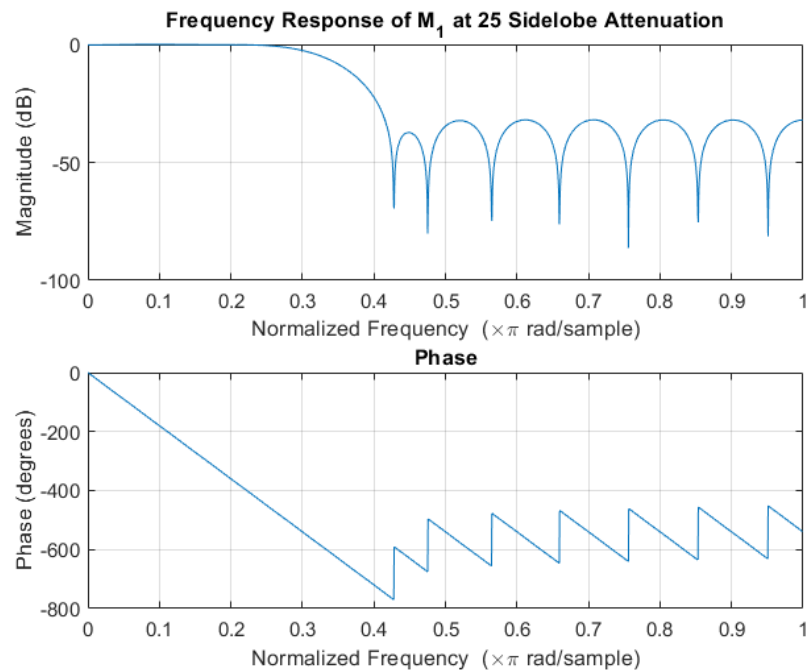


Figure 28: Frequency Response with order 20 and sidelobe attenuation of 25 dB for FIR filter

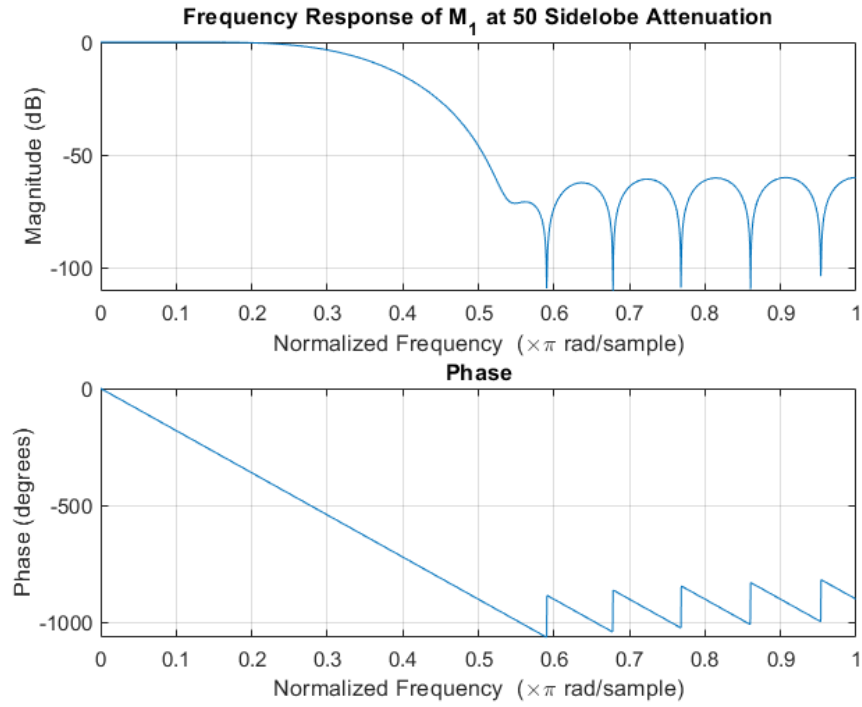


Figure 29: Frequency Response with order 20 and sidelobe attenuation of 50 dB for FIR Filter

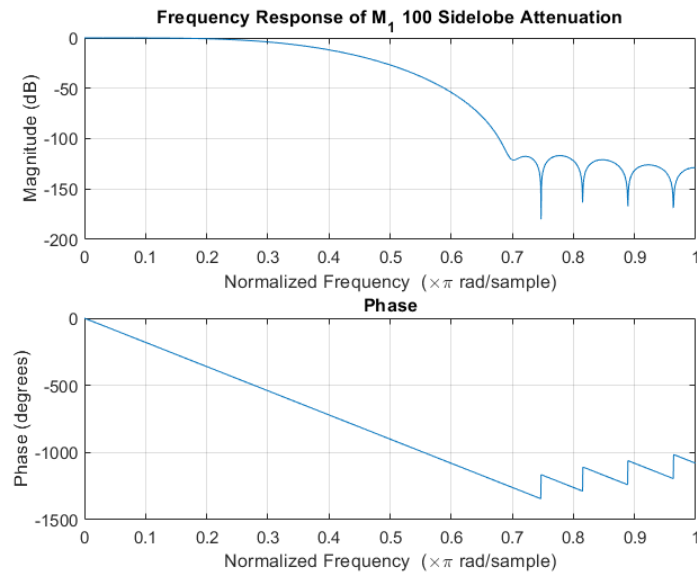


Figure 30: Frequency Response with order 20 and sidelobe attenuation of 100 dB for FIR Filter

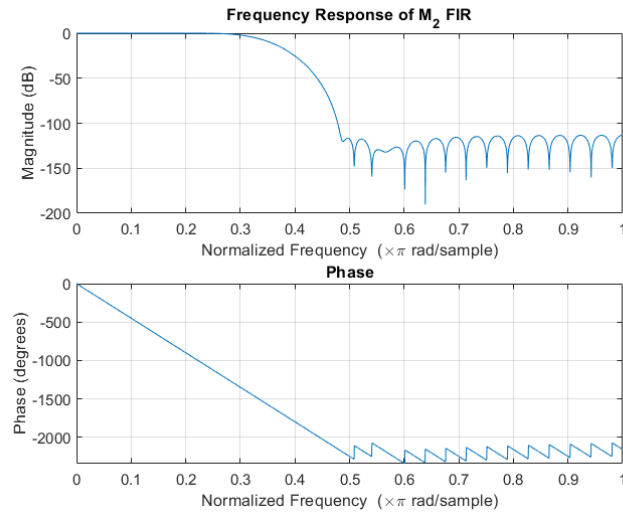


Figure 31: Frequency Response with order 50 and sidelobe attenuation of 100 dB for FIR Filter

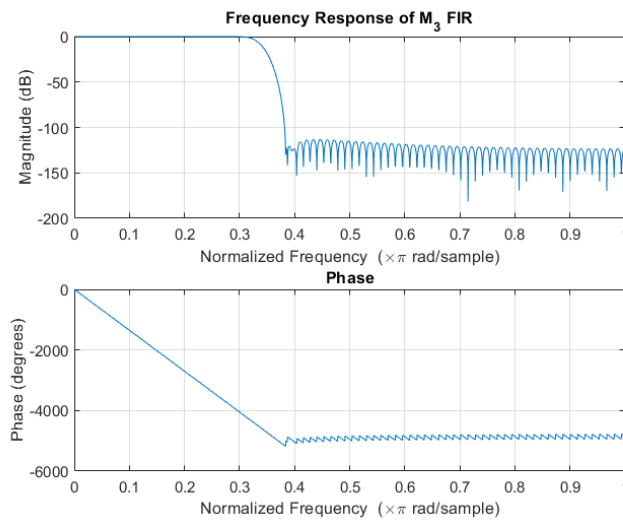


Figure 32: Frequency Response with order 150 and sidelobe attenuation of 100 dB for FIR Filter

2(b) Design Chebyshev Type II lowpass IIR filters with the cutoff frequency $\omega_c = \pi/3$ using the MATLAB Signal Processing Toolbox function `cheby2()` for filter orders $M = N = 20, 50$ and 150 . Note that you will

need to specify the stopband attenuation – make it the same as one of the values that you used for the Dolph-Chebyshev window sidelobe attenuation in part (a).

Answer and Discussion:

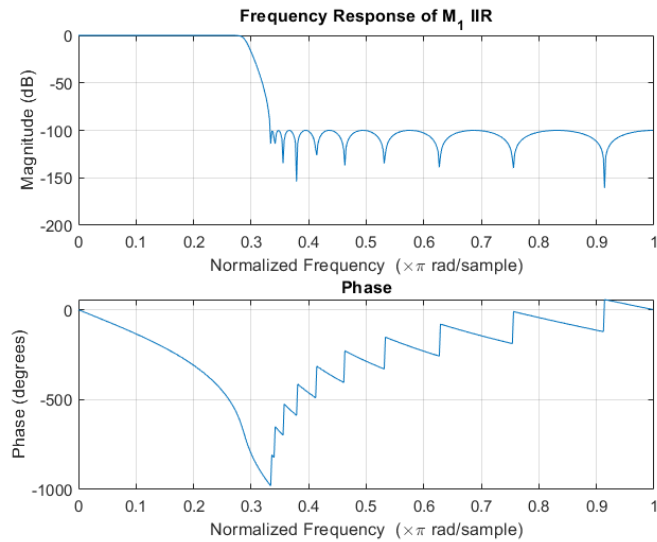


Figure 33: Frequency Response with order 20 and sidelobe attenuation of 100 dB for IIR Filter

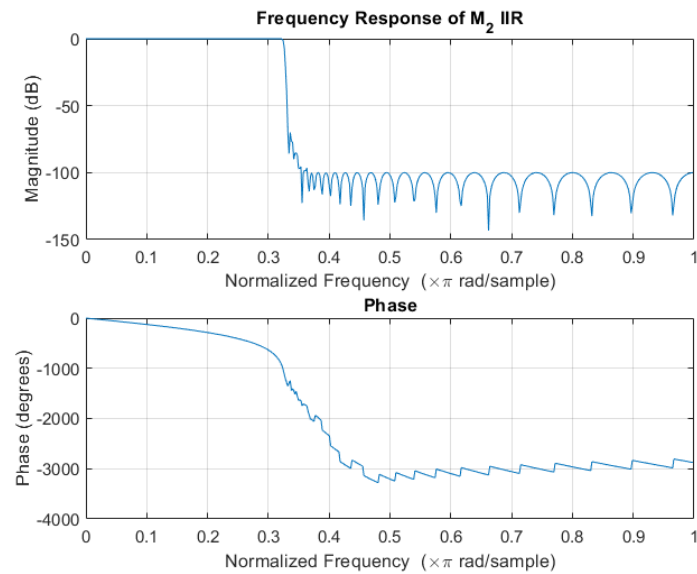


Figure 34: Frequency Response with order 50 and sidelobe attenuation of 100 dB for IIR Filter

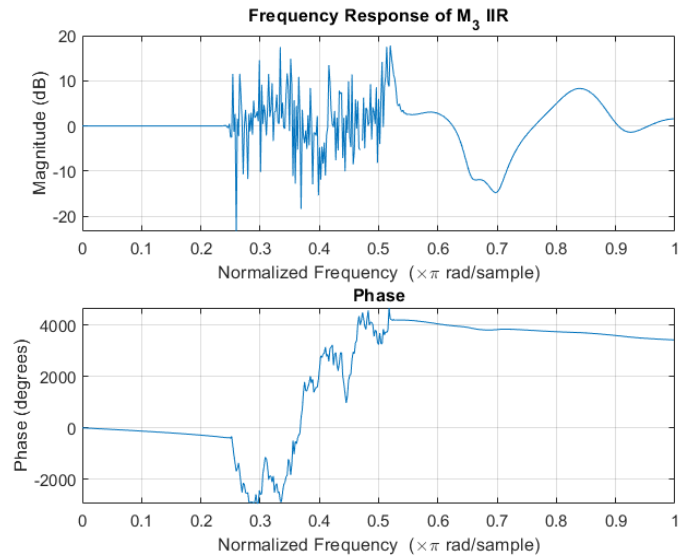


Figure 35: Frequency Response with order 150 and sidelobe attenuation of 100 dB IIR Filter

2(c) Determine the impulse responses of these filters. Do they appear to be infinite?

Answer and Discussion:

The impulse responses of order 20 and order 50 IIR filters shown in Figures 36 and 37 appear to converge to 0 as i approaches infinity. The impulse response of order 150 as shown in Figure 38 does not converge to 0, instead it exhibits unstable oscillatory behavior that exponentially grows in size as approaching infinity. This is because the higher order causes the poles to move close to or outside the unit circle.

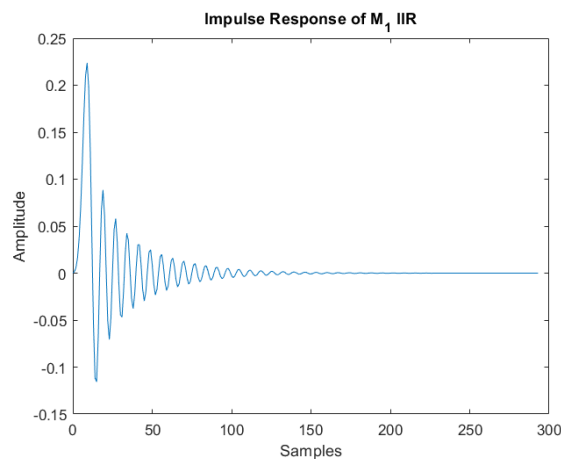


Figure 36: Impulse response of IIR filter of order 20

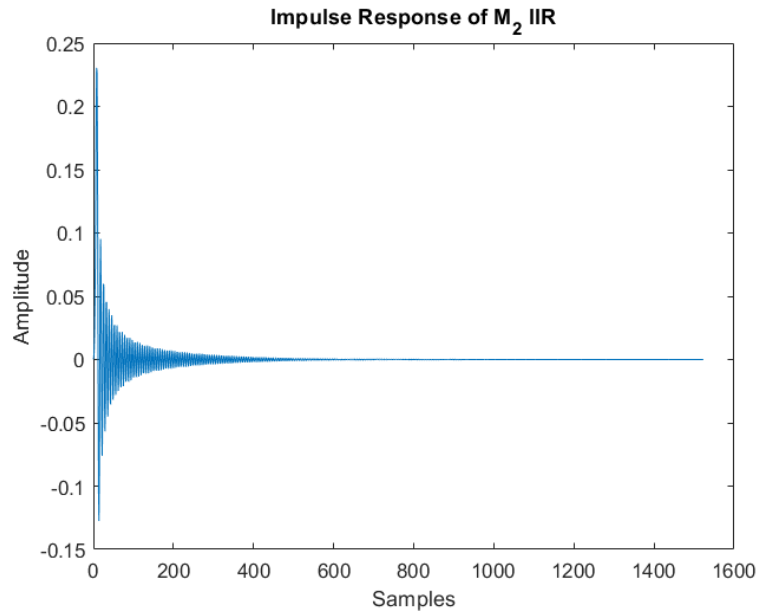


Figure 37: Impulse response of IIR filter of order 50

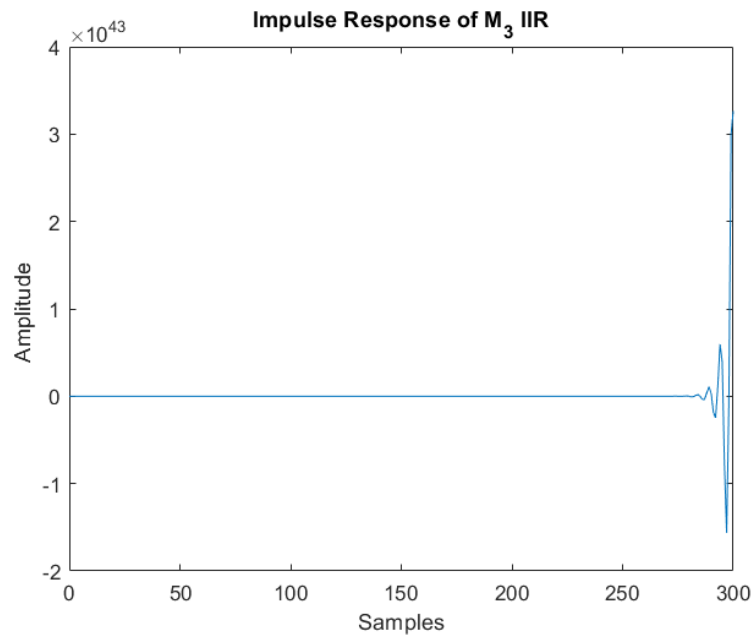


Figure 38: Impulse response of IIR filter of order 150

2(d) Compare and contrast the IIR filter frequency responses (both the magnitude and the phase) with the FIR frequency responses from part (a). Explain the differences in light of filter theory as it relates to causal FIR and IIR filter

Answer and Discussion:

Comparing Figure 34 and Figure 31 we see that the IIR filter has a steeper transition slope when moving from the pass band to the stop band. It also exhibits this behavior closer to the actual desired cut off. However, as a trade-off, we see that the IIR has more ripple, extending closer to the stop band. Additionally, all FIR filters were stable. Figure 35 shows that at a high enough order the IIR filter began exhibiting unstable behavior and therefore would not be adequate for real life filter implementation at that order.

2(e) Create a signal 1000 samples long of white Gaussian noise. Using the `filter()` function, calculate the output of the IIR filter for this signal. Now estimate the output by convolving the input signal with a very long impulse response that you obtained from part (c). How good is the estimate?

Answer and Discussion:

When using the `conv()` function with the very long impulse response we see that we match filter behavior but lose a portion of the signal as the impulse eventually goes to zero as approaching infinity. However, using `filter()` we get the full behavior expected from the filter even when using the impulse response. It should be noted with the unstable response of the order 150 IIR filter we do not get proper filtering and instead lose all the information contained with the filter. The termination when using `conv()` vs the full response when using `filter()` can be highlighted when looking at Figures 41 and 43.

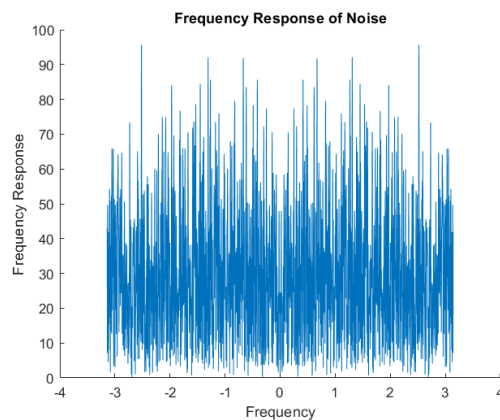


Figure 39: Frequency spectrum of unfiltered Gaussian Noise

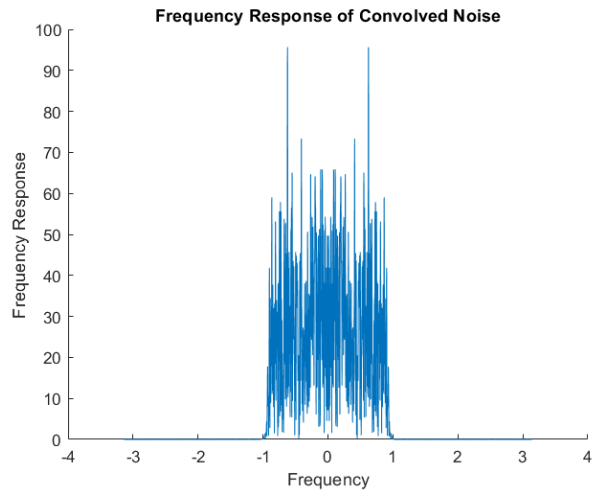


Figure 40: Frequency response when convolved with impulse response seen in Figure 36 which is with order 20 IIR filter using conv()

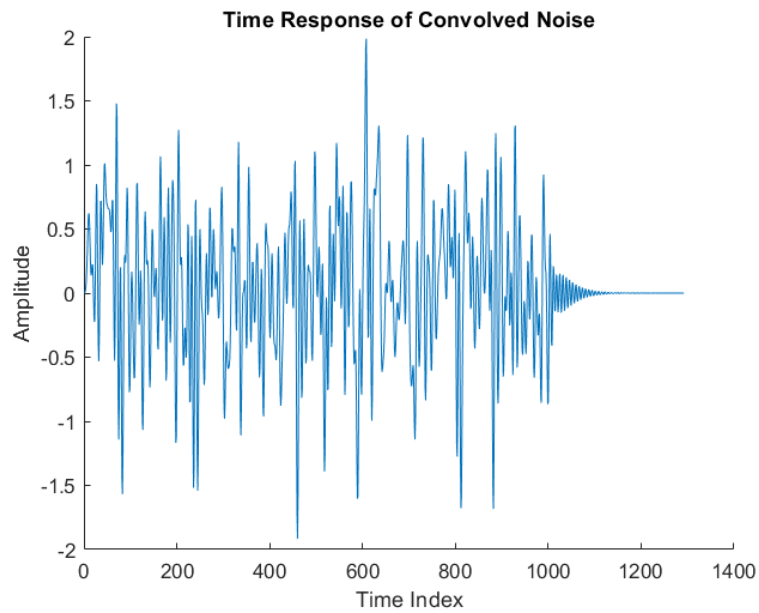


Figure 41: Time response when convolved with impulse response seen in Figure 36 which is with order 20 IIR filter using conv()

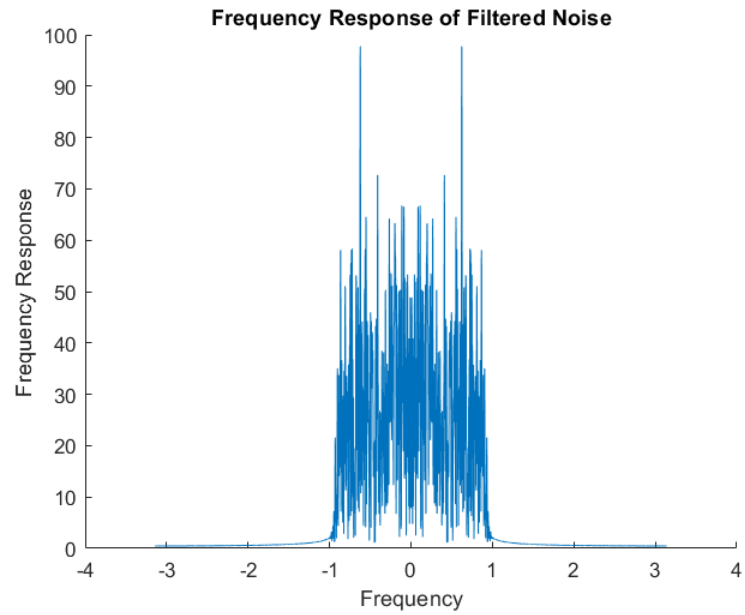


Figure 42: Frequency response when filtered with impulse response seen in Figure 36 which is with order 20 IIR filter using filter())

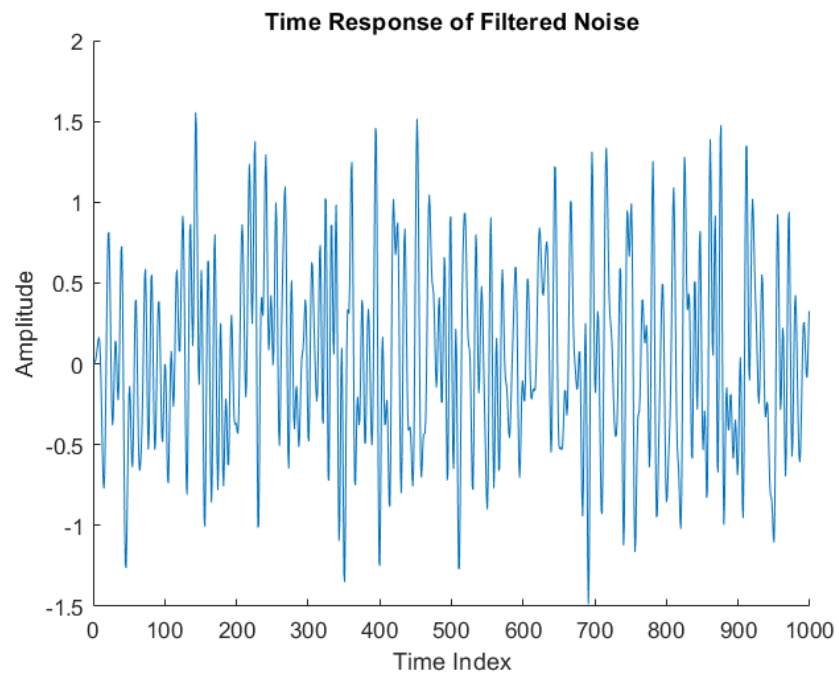


Figure 43: Time response when filtered with impulse response seen in Figure 36 which is with order 20 IIR filter using filter())

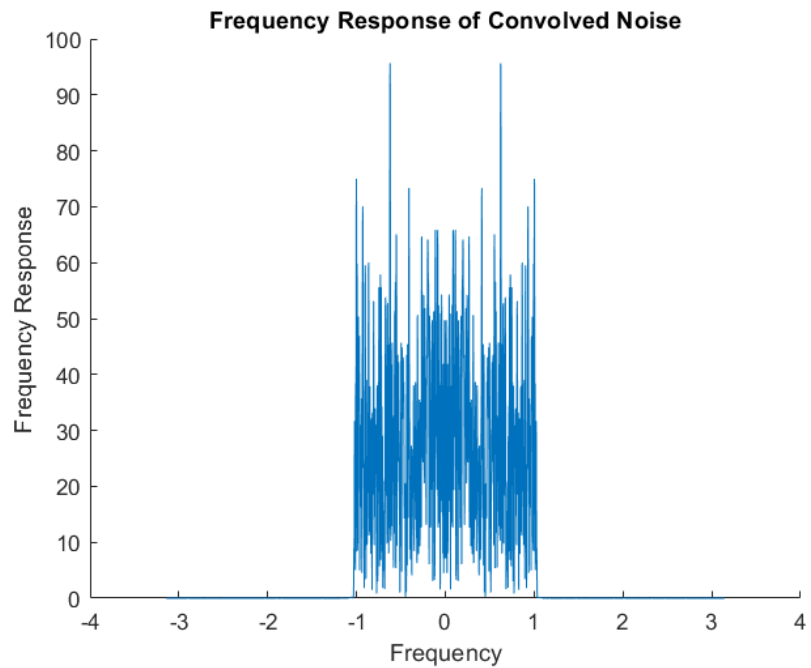


Figure 44: Frequency response when convolved with impulse response seen in Figure 37 which is with order 50 IIR conv()

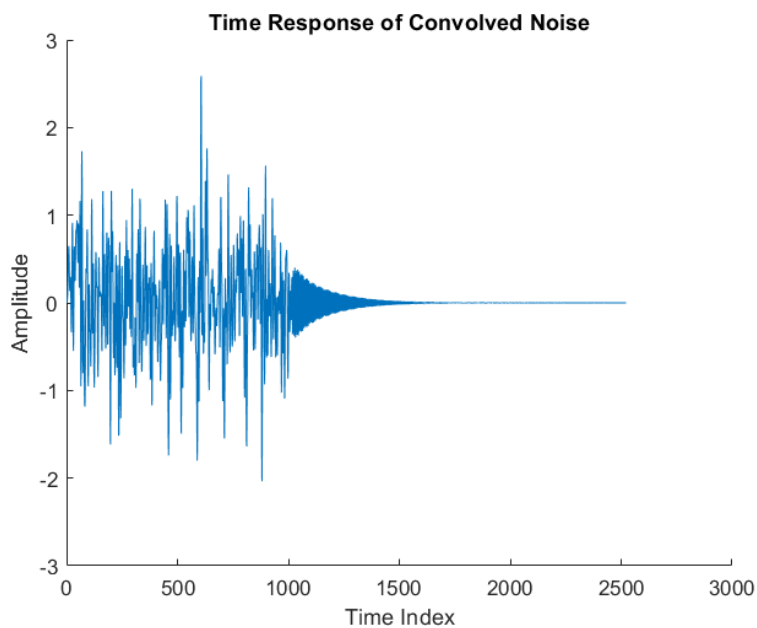


Figure 45: Time response when convolved with impulse response seen in Figure 37 which is with order 50 IIR conv()

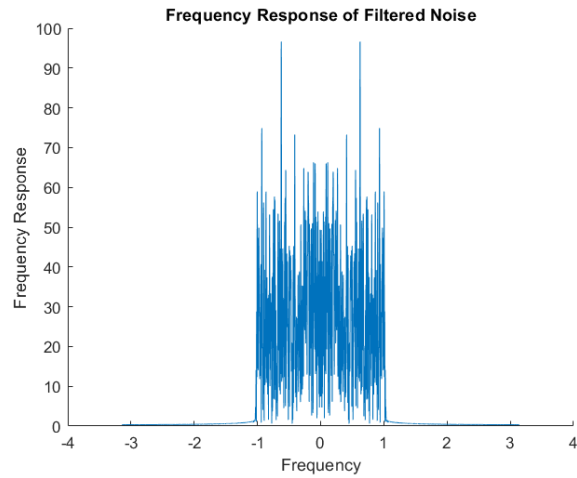


Figure 46: Frequency response when filtered with impulse response seen in Figure 37 which is with order 50 IIR filter()

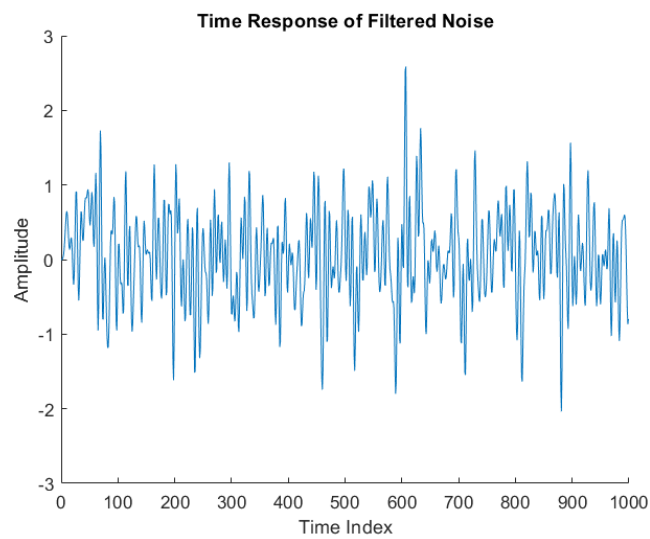


Figure 47: Time response when filtered with impulse response seen in Figure 37 which is with order 50 IIR filter()

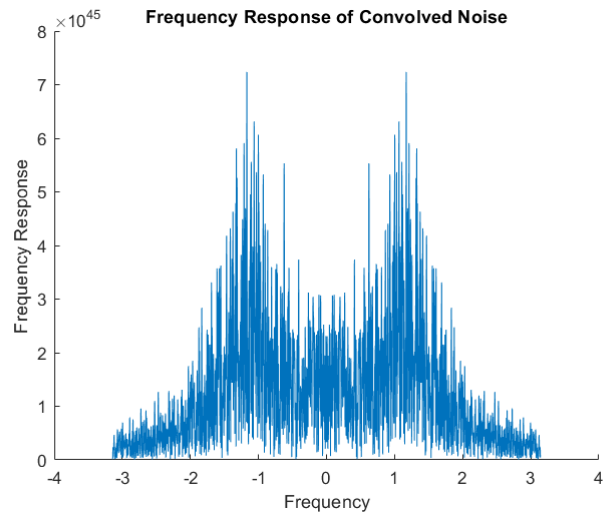


Figure 48: Frequency response when convolved with impulse response seen in Figure 38 which is with order 150 IIR conv()

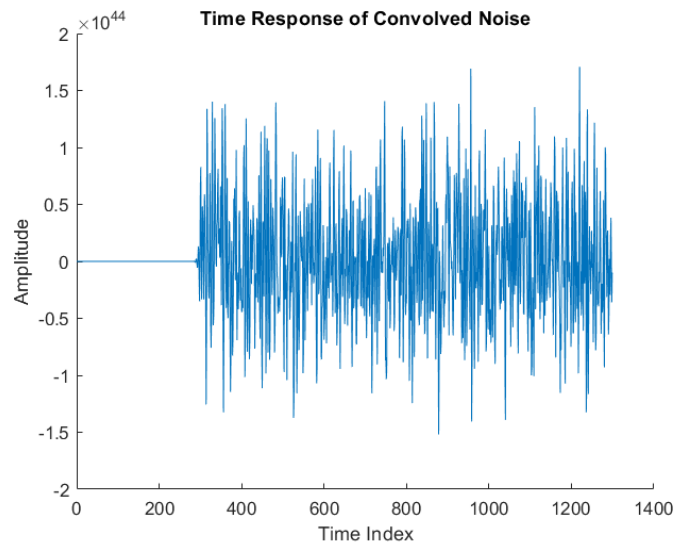


Figure 49: Time response when convolved with impulse response seen in Figure 38 which is with order 150 IIR conv()

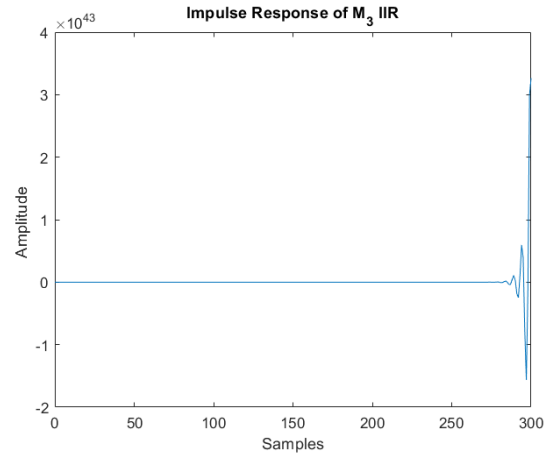


Figure 50: Impulse response of order 150 IIR filter

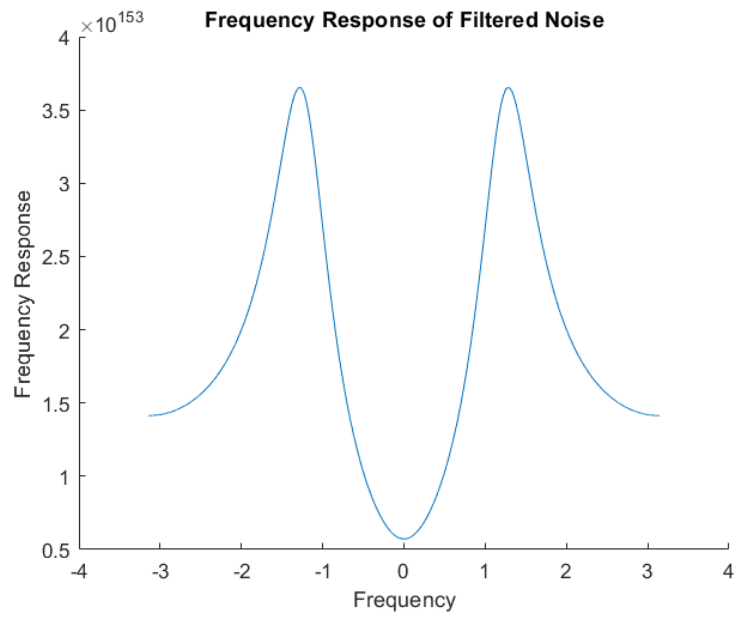


Figure 51: Frequency response when filtered with impulse response seen in Figure 38 which is with order 150 IIR filter()

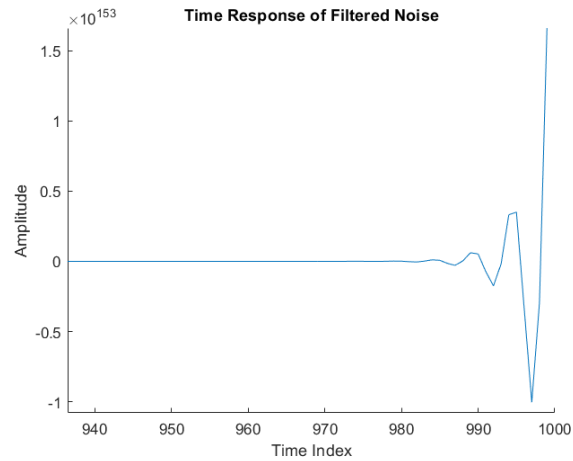


Figure 52: Time response when filtered with impulse response seen in Figure 38 which is with order 150 IIR filter()

3. Detecting Targets in Background Clutter Utilizing a Moving Target Indicator with the TMS 320 DSP Processor

This section focuses on the Moving Target Indicator (MTI) applied to radar data to filter out static clutter (barriers) using the TMS 320 DSP platform.

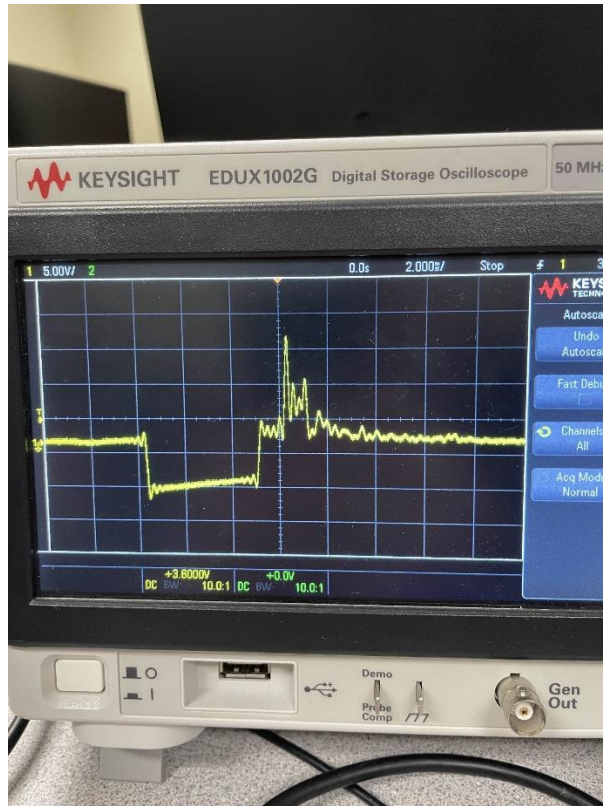


Figure 53: Range plot of the signal from lab 4 but with addition of barrier

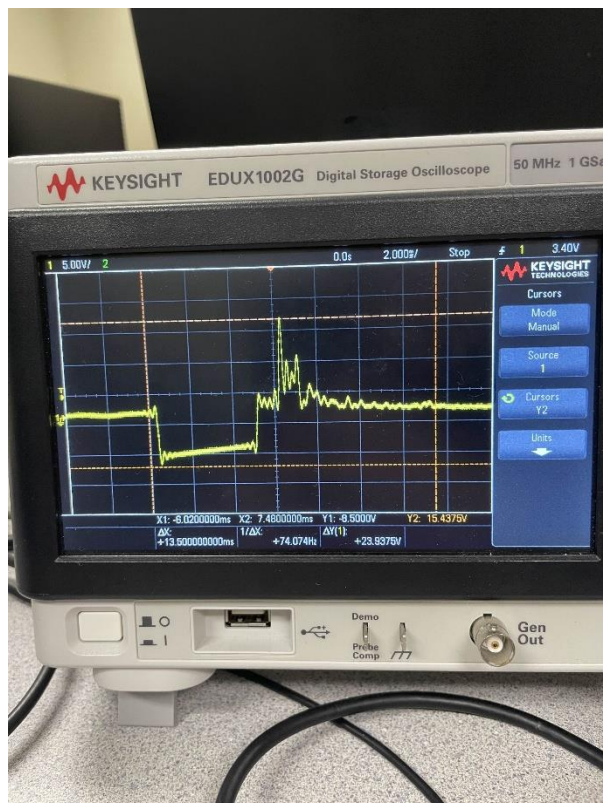


Figure 54: Range plot of the signal with barrier and without Moving Target Indicator processing

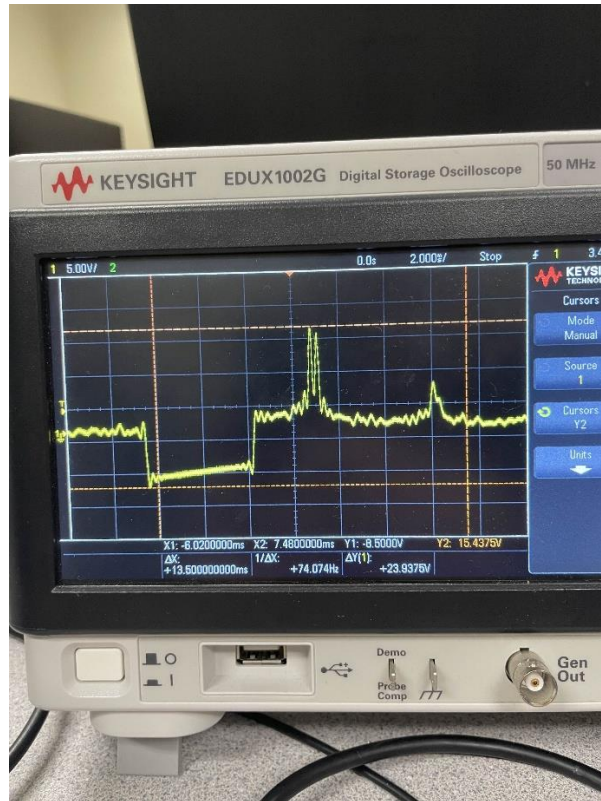


Figure 55: Range plot of the signal with barrier and with Two-Pulse Cancellor Moving Target Indicator processing

The range plot without MTI returns a signal that mixes the true target return signal with barrier return signals. As a result, it is impossible to distinguish between the two. The range plot with MTI filters out the signals that comes from the barrier and leaving just the true targets' signals. As we can observe in the above two figures, the signal without MTI shows additional pulses that is not present in the signal with MTI. These additional pulses are from the barrier, and the two-pulse canceller MTI is able to filter out these signals, resulting in a signal of just the true targets.

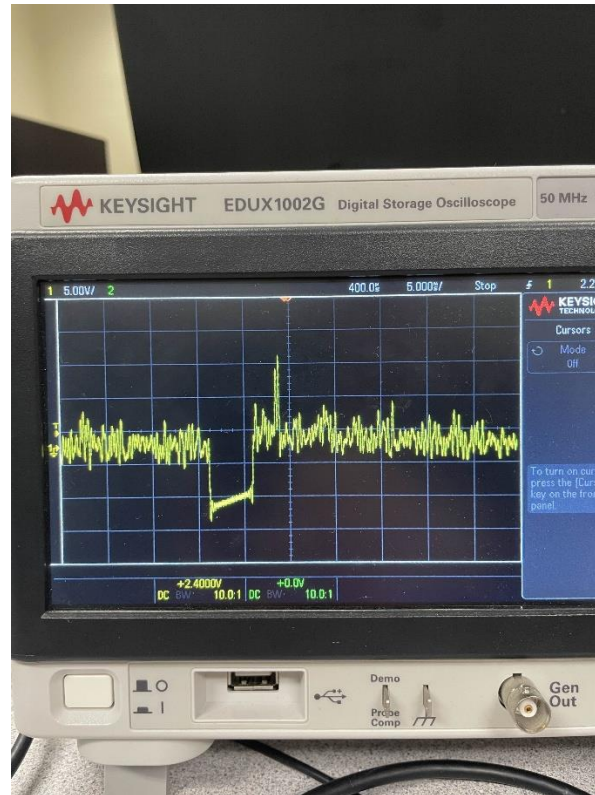


Figure 56: Range plot of the signal with barrier and with Three-Pulse Cancellor Moving Target Indicator processing

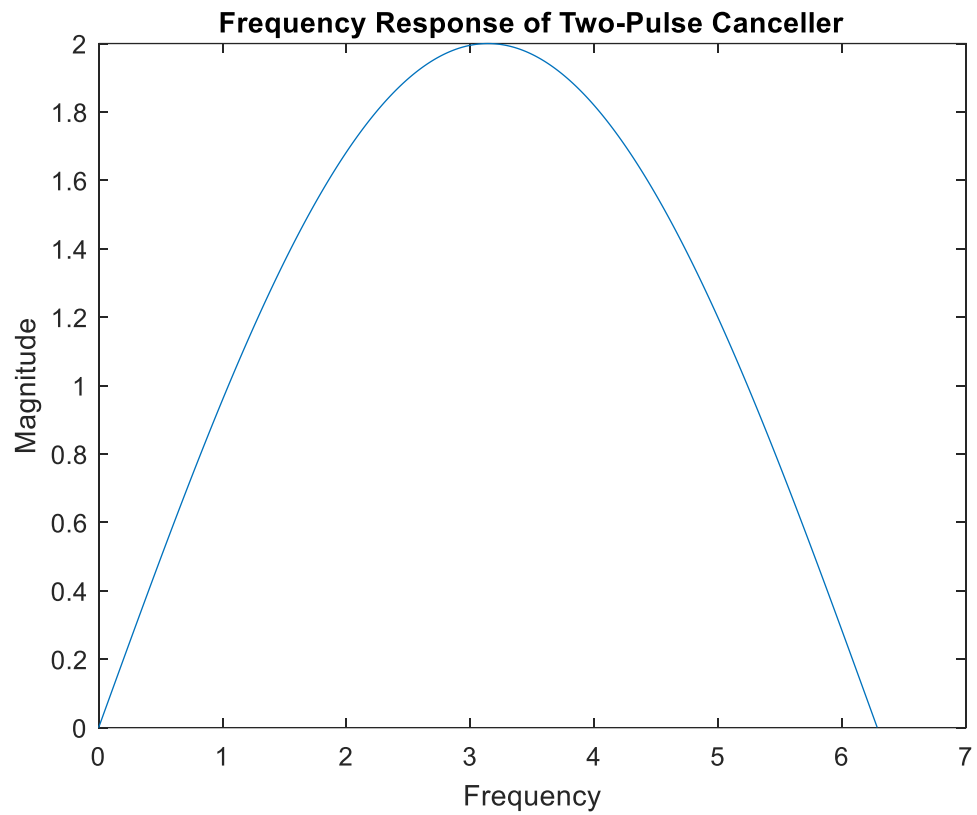


Figure 57: Frequency response of the two-pulse canceller MTI filter

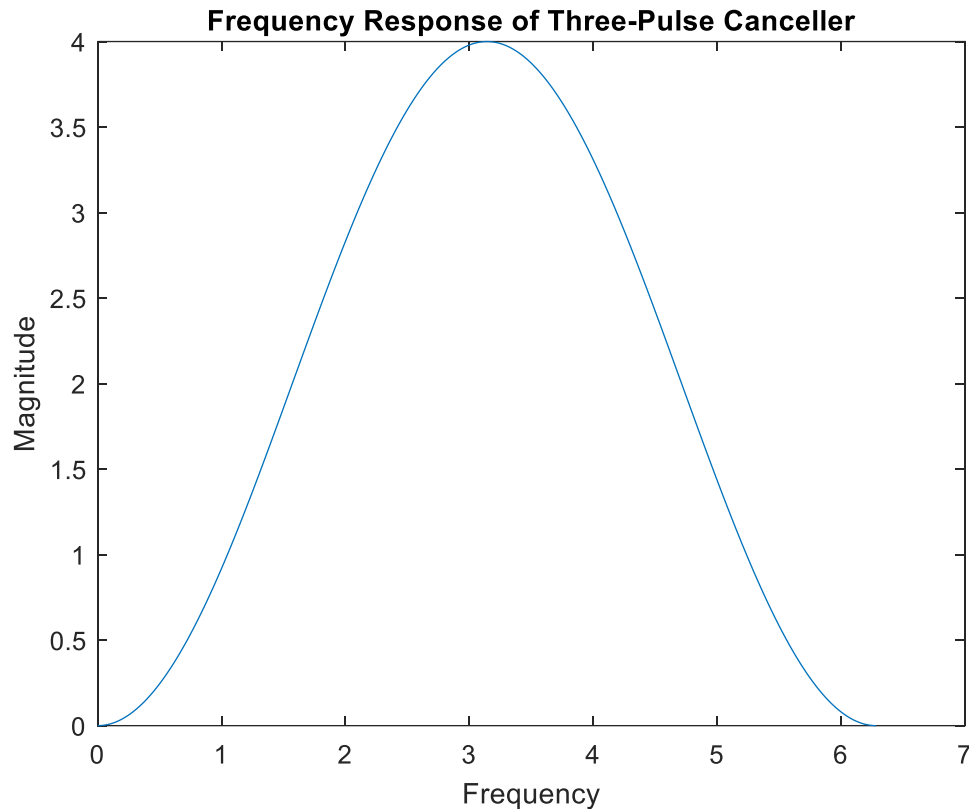


Figure 58: Frequency response of the three-pulse canceller MTI filter

In our case, where our target is slow moving, it is not more effective to use the three-pulse canceller MTI compared to the two-pulse canceller MTI. This is because slower moving targets have a lower Doppler frequency, and as we can observe in the above two figures, the two-pulse canceller filter rises in magnitude much faster at lower frequencies compared to the three-pulse canceller filter. What this means is that in the three-pulse canceller filter, our target's signal will fall in a region where the filter's magnitude is much lower in relation to the other regions of the filter. As a result, any noise components that fall in the other regions will be amplified much higher in relation to the target's signal. Compared to the two-pulse canceller where the filter's magnitude rises faster at lower frequencies, other regions where the noise components fall in may still be amplified in relation to the region where our target falls into, however, it will be at a much lower multiple.

When we verify this with the oscilloscope observation, it is clear that the three-pulse canceller MTI filter results in a significantly noisier signal compared to the two-pulse filter. In addition, we can see that the far-right pulse that can be seen with the two-pulse canceller filter becomes indistinguishable from the noise with the three-pulse canceller filter.

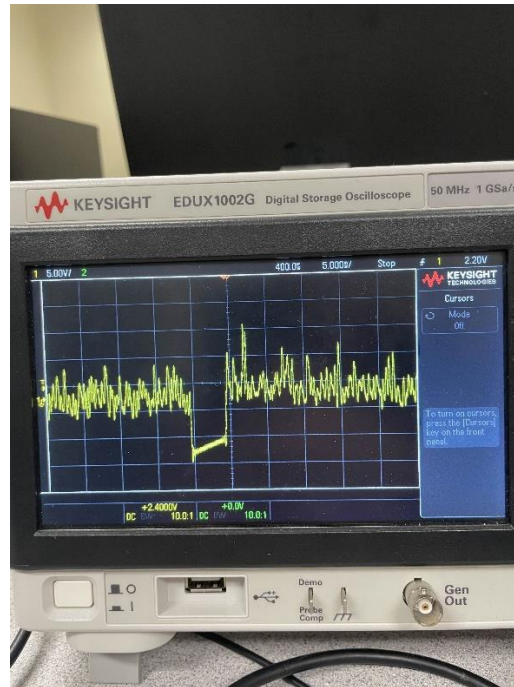


Figure 59: Range plot of the signal with barrier and with Three-Pulse Canceller Moving Target Indicator processing, with an added Hanning window for FFT calculations

If the platform where the radar is on is moving instead of stationary, this means that the Doppler shift of the barrier will no longer be zero. This is because the Doppler shift is proportional to the velocity difference between the object and the platform, and with a stationary barrier and moving platform, this difference is non-zero. In this case, we want to use a band-stop filter, where the Doppler frequency of the barrier falls within the stop band, and our targets' Doppler frequencies fall within the pass-band.